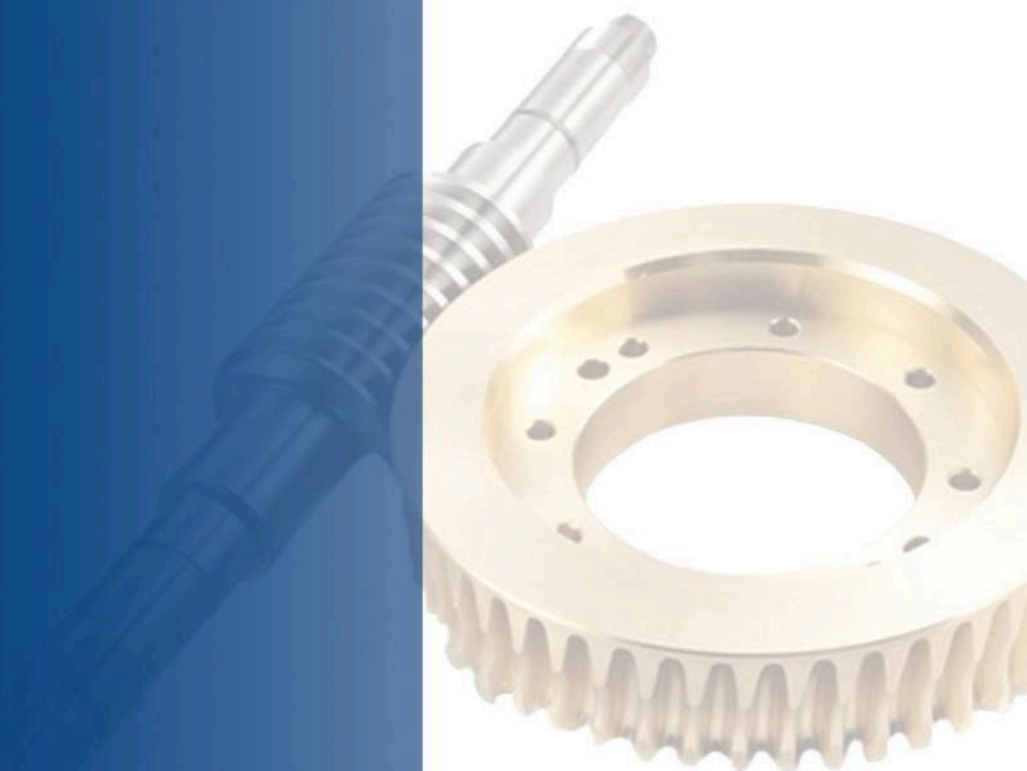
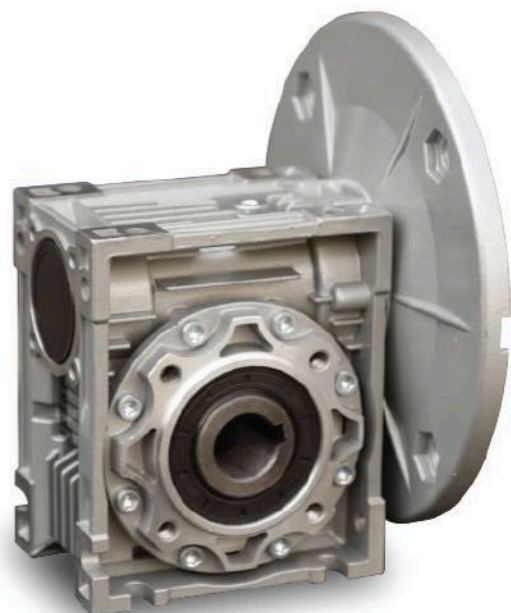


# ALUMINIUM

## WORM GEAR REDUCER SPECIFICATION WORM GEAR

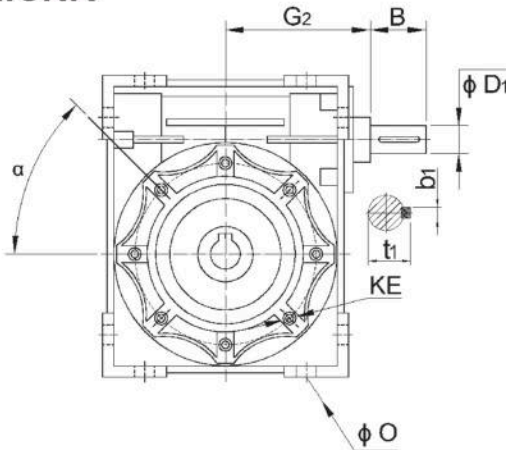


ISO 9001  
IEC  
CE

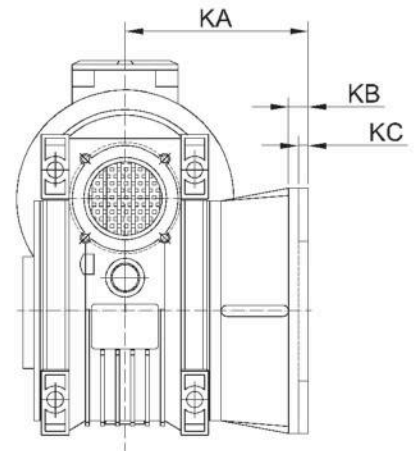
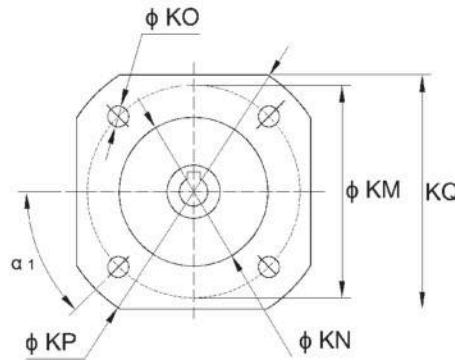
**Max Loyd**  
moving power

## Single MCN series installation dimensions

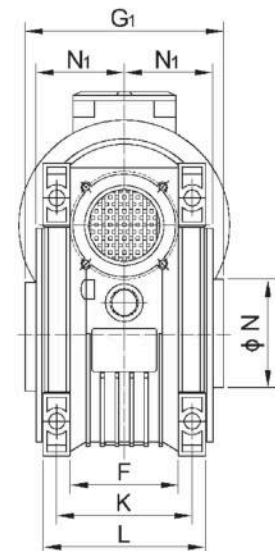
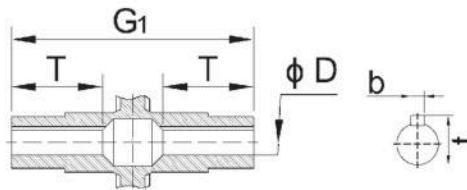
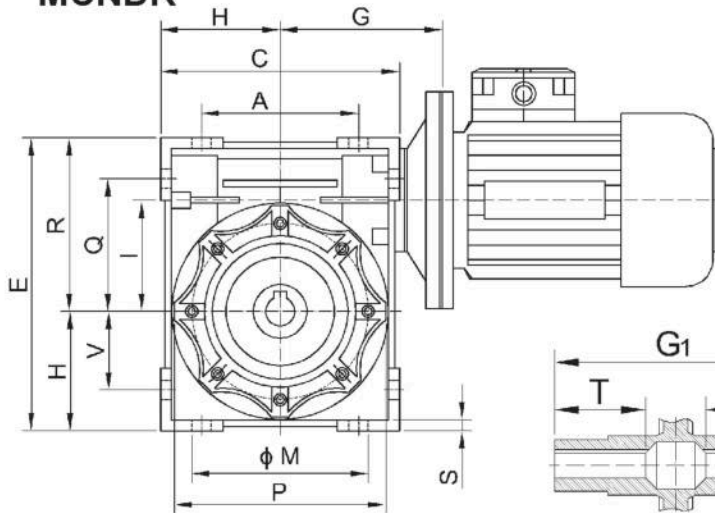
### MCNK



### MCNDKO



### MCNDK



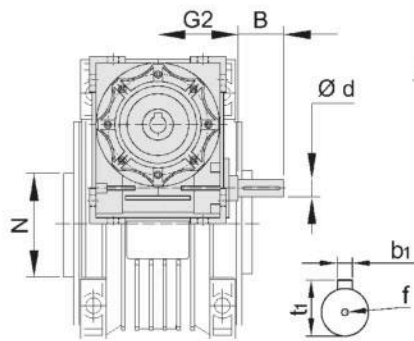
| Size | A   | B  | C     | D (H7)  | D <sub>1</sub> (J6) | E     | F  | G     | G <sub>1</sub> | G <sub>2</sub> | H     | I   | L   | M   | N (h8) | N <sub>1</sub> | O   | P   | Q   | R     |
|------|-----|----|-------|---------|---------------------|-------|----|-------|----------------|----------------|-------|-----|-----|-----|--------|----------------|-----|-----|-----|-------|
| 30   | 54  | 20 | 80    | 14      | 9                   | 97    | 32 | 55    | 63             | 51             | 40    | 30  | 56  | 65  | 55     | 29             | 6.5 | 75  | 44  | 57    |
| 40   | 70  | 23 | 100   | 18 (19) | 11                  | 121.5 | 43 | 70    | 78             | 60             | 50    | 40  | 71  | 75  | 60     | 36.5           | 6.5 | 87  | 55  | 71.5  |
| 50   | 80  | 30 | 120   | 25 (24) | 14                  | 144   | 49 | 80    | 92             | 74             | 60    | 50  | 85  | 85  | 70     | 43.5           | 8.5 | 100 | 64  | 84    |
| 63   | 100 | 40 | 144   | 25 (28) | 19                  | 174   | 67 | 95    | 112            | 90             | 72    | 63  | 103 | 95  | 80     | 53             | 8.5 | 110 | 80  | 102   |
| 75   | 120 | 50 | 172   | 28 (35) | 24                  | 205   | 72 | 112.5 | 120            | 105            | 86    | 75  | 112 | 115 | 95     | 57             | 11  | 140 | 93  | 119   |
| 90   | 140 | 50 | 208   | 35 (38) | 24                  | 238   | 74 | 129.5 | 140            | 125            | 103   | 90  | 130 | 130 | 110    | 67             | 13  | 160 | 102 | 135   |
| 110  | 170 | 60 | 252.5 | 42      | 28                  | 295   | -  | 160   | 155            | 142            | 127.5 | 110 | 144 | 165 | 130    | 74             | 14  | 200 | 125 | 167.5 |
| 130  | 200 | 80 | 292.5 | 45      | 30                  | 335   | -  | 180   | 170            | 162            | 147.5 | 130 | 155 | 215 | 180    | 81             | 16  | 250 | 140 | 187.5 |
| 150  | 240 | 80 | 340   | 50      | 35                  | 400   | -  | 210   | 200            | 192            | 170   | 150 | 185 | 215 | 180    | 96             | 18  | 250 | 180 | 230   |

| Size | S   | T    | V   | K   | KA   |      |     | KB |    |    | KC |    |    | KE          | α   | α <sub>1</sub> | KM  |     |     | KN (H8) |     |     | KO       |          |         | KP  |     |     | KQ  |    |     | b      | b <sub>1</sub> | f   | t          | t <sub>1</sub> | kg  |
|------|-----|------|-----|-----|------|------|-----|----|----|----|----|----|----|-------------|-----|----------------|-----|-----|-----|---------|-----|-----|----------|----------|---------|-----|-----|-----|-----|----|-----|--------|----------------|-----|------------|----------------|-----|
|      |     |      |     |     | F    | FB   | FL  | F  | FB | FL | F  | FB | FL |             |     |                | F   | FB  | FL  | F       | FB  | FL  | F        | FB       | FL      | F   | FB  | FL  | F   | FB | FL  |        |                |     |            |                |     |
| 30   | 5.5 | 21   | 27  | 44  | 54.5 | -    | -   | 6  | -  | -  | 4  | -  | -  | M6X11(n,4)  | 0°  | 45°            | 68  | -   | -   | 50      | -   | -   | 6.5(n,4) | -        | -       | 80  | -   | -   | 70  | -  | -   | 5      | 3              | -   | 16.3       | 10.2           | 1.2 |
| 40   | 6.5 | 26   | 35  | 60  | 67   | 76.5 | 97  | 7  | 9  | 7  | 4  | 5  | 4  | M6X8(n,4)   | 45° | 45°            | 87  | 115 | 87  | 60      | 95  | 60  | 9(n,4)   | 9.5(n,4) | 9(n,4)  | 110 | 140 | 110 | 95  | -  | 95  | 6(6)   | 4              | -   | 20.8(21.8) | 12.5           | 2.3 |
| 50   | 7   | 30   | 40  | 70  | 90   | 87.5 | 120 | 9  | 10 | 9  | 5  | 5  | 5  | M8X10(n,4)  | 45° | 45°            | 90  | 130 | 90  | 70      | 110 | 70  | 11(n,4)  | 9.5(n,4) | 11(n,4) | 125 | 160 | 125 | 110 | -  | 110 | 8(8)   | 5              | M6  | 28.3(27.3) | 16.0           | 3.5 |
| 63   | 8   | 36   | 50  | 85  | 82   | 99   | 112 | 10 | 11 | 10 | 6  | 5  | 6  | M8X14(n,8)  | 45° | 45°            | 150 | 165 | 150 | 115     | 130 | 115 | 11(n,4)  | 11(n,4)  | 11(n,4) | 180 | 200 | 180 | 142 | -  | 142 | 8(8)   | 6              | M6  | 28.3(31.3) | 21.5           | 6.2 |
| 75   | 10  | 40   | 60  | 90  | 111  | -    | -   | 13 | -  | -  | 6  | -  | -  | M8X14(n,8)  | 45° | 45°            | 165 | -   | -   | 130     | -   | -   | 14(n,4)  | -        | -       | 200 | -   | -   | 170 | -  | -   | 8(10)  | 8              | M8  | 31.3(38.3) | 27.0           | 9   |
| 90   | 11  | 45   | 70  | 100 | 111  | -    | -   | 13 | -  | -  | 6  | -  | -  | M10X18(n,8) | 45° | 45°            | 175 | -   | -   | 152     | -   | -   | 14(n,4)  | -        | -       | 210 | -   | -   | 200 | -  | -   | 10(10) | 8              | M8  | 38.3(41.3) | 27.0           | 13  |
| 110  | 14  | 50   | 85  | 115 | 131  | -    | -   | 15 | -  | -  | 6  | -  | -  | M10X18(n,8) | 45° | 45°            | 230 | -   | -   | 170     | -   | -   | 14(n,4)  | -        | -       | 280 | -   | -   | 260 | -  | -   | 12     | 8              | M10 | 45.3       | 31.0           | 35  |
| 130  | 15  | 60   | 100 | 120 | 140  | -    | -   | 15 | -  | -  | 6  | -  | -  | M12X21(n,8) | 45° | 22.5°          | 255 | -   | -   | 180     | -   | -   | 16(n,8)  | -        | -       | 320 | -   | -   | 290 | -  | -   | 14     | 8              | M10 | 48.8       | 33.0           | 48  |
| 150  | 18  | 72.5 | 120 | 145 | 155  | -    | -   | 15 | -  | -  | 6  | -  | -  | M12X21(n,8) | 45° | 22.5°          | 255 | -   | -   | 180     | -   | -   | 16(n,8)  | -        | -       | 320 | -   | -   | 290 | -  | -   | 14     | 10             | M12 | 53.8       | 38.0           | 84  |

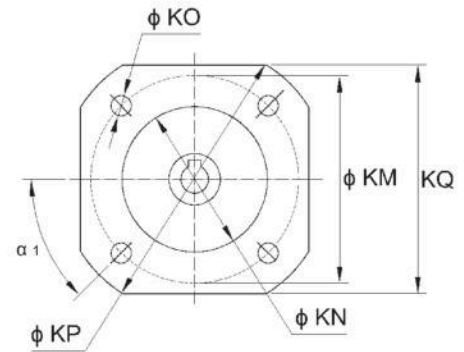
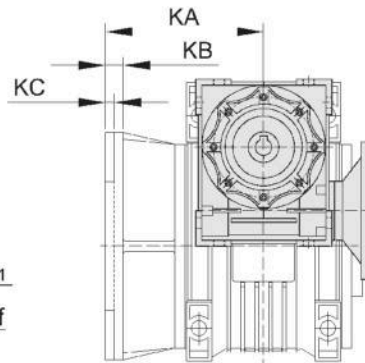


## Double MCEN ( worm-worm ) series installation dimensions

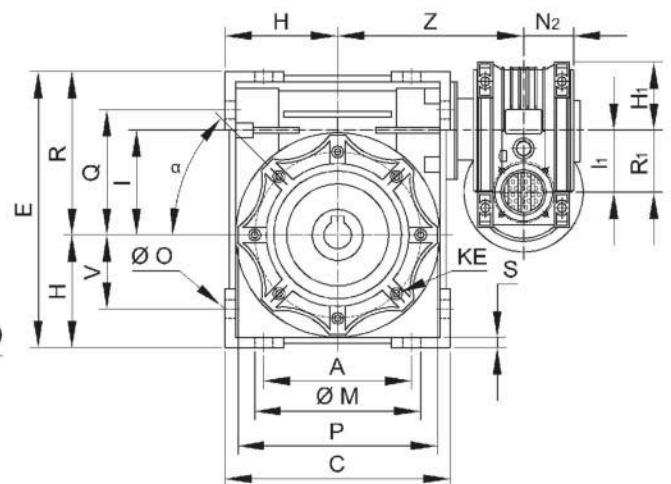
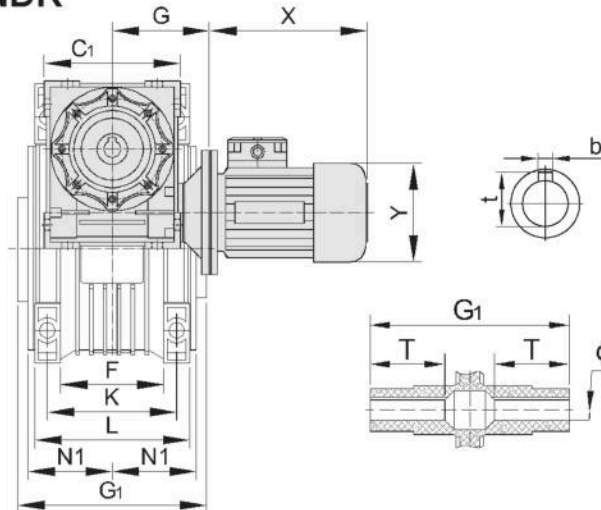
### MCENK



### MCENDKO



### MCENDK

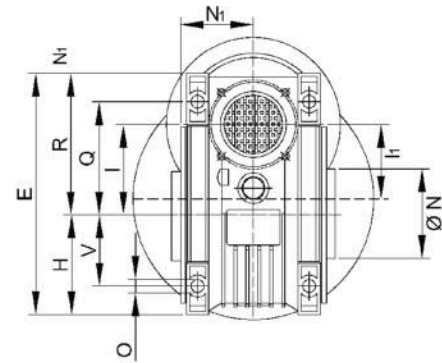
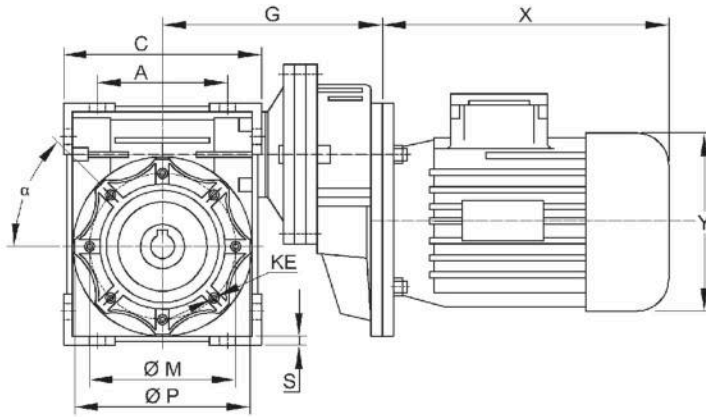


| Size   | A   | B  | C     | C1  | D(H7)  | d (j6) | E     | F  | G  | G1  | G2 | H     | H1 | I   | I1 | L   | M   | N(h8) | N1   | N2   | O   | P   | Q   | R     | R1   | S   | T    | V   | Z   | K   | KA   |      |     |
|--------|-----|----|-------|-----|--------|--------|-------|----|----|-----|----|-------|----|-----|----|-----|-----|-------|------|------|-----|-----|-----|-------|------|-----|------|-----|-----|-----|------|------|-----|
|        |     |    |       |     |        |        |       |    |    |     |    |       |    |     |    |     |     |       |      |      |     |     |     |       |      |     |      |     |     |     | F    | FB   | FL  |
| 25/30  | 54  | -  | 80    | 70  | 14     | -      | 97    | 32 | 45 | 63  | -  | 40    | 35 | 30  | 25 | 56  | 65  | 55    | 29   | 22.5 | 6.5 | 75  | 44  | 57    | 48   | 5.5 | 21   | 27  | 100 | 44  | 54.5 | -    | -   |
| 25/40  | 70  | -  | 100   | 70  | 18(19) | -      | 121.5 | 43 | 45 | 78  | -  | 50    | 35 | 40  | 25 | 71  | 75  | 60    | 36.5 | 22.5 | 6.5 | 87  | 55  | 71.5  | 48   | 6.5 | 26   | 35  | 115 | 60  | 67   | 76.5 | 97  |
| 30/40  | 70  | 20 | 100   | 80  | 18(19) | 9      | 121.5 | 43 | 55 | 78  | 51 | 50    | 40 | 40  | 30 | 71  | 75  | 60    | 36.5 | 29   | 6.5 | 87  | 55  | 71.5  | 57   | 6.5 | 26   | 35  | 120 | 60  | 67   | 76.5 | 97  |
| 30/50  | 80  | 20 | 120   | 80  | 25(24) | 9      | 144   | 49 | 55 | 92  | 51 | 60    | 40 | 50  | 30 | 85  | 85  | 70    | 43.5 | 29   | 8.5 | 100 | 64  | 84    | 57   | 7   | 30   | 40  | 130 | 70  | 90   | 87.5 | 120 |
| 30/63  | 100 | 20 | 144   | 80  | 25(28) | 9      | 174   | 67 | 55 | 112 | 51 | 72    | 40 | 63  | 30 | 103 | 95  | 80    | 53   | 29   | 8.5 | 110 | 80  | 102   | 57   | 8   | 36   | 50  | 145 | 85  | 82   | 99   | 112 |
| 40/75  | 120 | 23 | 172   | 100 | 28(35) | 11     | 205   | 72 | 70 | 120 | 60 | 86    | 50 | 75  | 40 | 112 | 115 | 95    | 57   | 36.5 | 11  | 140 | 93  | 119   | 71.5 | 10  | 40   | 60  | 165 | 90  | 111  | -    | -   |
| 40/90  | 140 | 23 | 208   | 100 | 35(38) | 11     | 238   | 74 | 70 | 140 | 60 | 103   | 50 | 90  | 40 | 130 | 130 | 110   | 67   | 36.5 | 13  | 160 | 102 | 135   | 71.5 | 11  | 45   | 70  | 182 | 100 | 111  | -    | -   |
| 50/110 | 170 | 30 | 252.5 | 120 | 42     | 14     | 295   | -  | 80 | 155 | 74 | 127.5 | 60 | 110 | 50 | 144 | 165 | 130   | 74   | 43.5 | 14  | 200 | 125 | 168   | 84   | 14  | 50   | 85  | 225 | 115 | 131  | -    | -   |
| 63/130 | 200 | 40 | 292.5 | 144 | 45     | 19     | 335   | -  | 95 | 170 | 90 | 147.5 | 72 | 130 | 63 | 155 | 215 | 180   | 81   | 53   | 16  | 250 | 140 | 187.5 | 102  | 15  | 60   | 100 | 245 | 120 | 140  | -    | -   |
| 63/150 | 240 | 40 | 340   | 144 | 50     | 19     | 400   | -  | 95 | 200 | 90 | 170   | 72 | 150 | 63 | 185 | 215 | 180   | 96   | 53   | 18  | 250 | 180 | 230   | 102  | 18  | 72.5 | 120 | 275 | 145 | 155  | -    | -   |

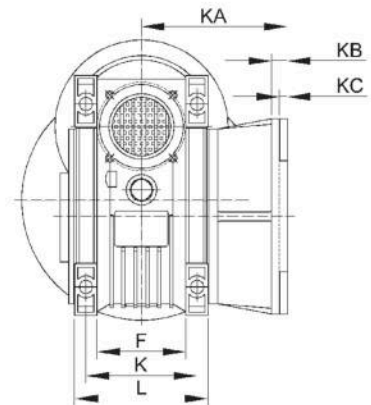
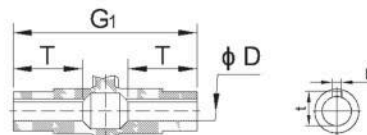
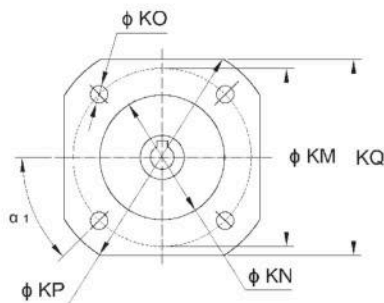
| KB |    |    | KC   | KE            | α   | α <sub>1</sub> | KM  |     |     | KN(H8) |     |     | KO       |          |         | KP  |     |     | KQ  |    |     | b      | b <sub>1</sub> | f  | t          | t <sub>1</sub> | kg   |
|----|----|----|------|---------------|-----|----------------|-----|-----|-----|--------|-----|-----|----------|----------|---------|-----|-----|-----|-----|----|-----|--------|----------------|----|------------|----------------|------|
| F  | FB | FL |      |               |     |                | F   | FB  | FL  | F      | FB  | FL  | F        | FB       | FL      | F   | FB  | FL  | F   | FB | FL  |        |                |    |            |                |      |
| 6  | -  | -  | 4    | M6 x 11(n,4)  | 0°  | 90°            | 68  | -   | -   | 50     | -   | -   | 6.5(n,4) | -        | -       | 80  | -   | -   | 70  | -  | -   | 5      | -              | -  | 16.3       | -              | 2.1  |
| 7  | 9  | 7  | 4(5) | M6 x 8(n,4)   | 45° | 90°            | 87  | 115 | 87  | 60     | 95  | 60  | 9(n,4)   | 9.5(n,4) | 9(n,4)  | 110 | 140 | 110 | 95  | -  | 95  | 6(6)   | -              | -  | 20.8(21.8) | -              | 3.2  |
| 7  | 9  | 7  | 4(5) | M6 x 8(n,4)   | 45° | 90°            | 87  | 115 | 87  | 60     | 95  | 60  | 9(n,4)   | 9.5(n,4) | 9(n,4)  | 110 | 140 | 110 | 110 | -  | 110 | 6(6)   | 3              | -  | 20.8(21.8) | 10.2           | 3.9  |
| 9  | 10 | 9  | 5(5) | M8 x 10(n,4)  | 45° | 90°            | 90  | 130 | 90  | 70     | 110 | 70  | 11(n,4)  | 9.5(n,4) | 11(n,4) | 125 | 160 | 125 | 110 | -  | 110 | 8(8)   | 3              | -  | 28.3(27.3) | 10.2           | 5.0  |
| 10 | 11 | 10 | 6(6) | M8 x 14(n,8)  | 45° | 90°            | 150 | 165 | 150 | 115    | 130 | 115 | 11(n,4)  | 11(n,4)  | 11(n,4) | 180 | 200 | 180 | 142 | -  | 142 | 8(8)   | 3              | -  | 28.3(31.3) | 10.2           | 7.8  |
| 13 | -  | -  | 6    | M8 x 14(n,8)  | 45° | 90°            | 165 | -   | -   | 130    | -   | -   | 14(n,4)  | -        | -       | 200 | -   | -   | 170 | -  | -   | 8(10)  | 4              | -  | 31.3(38.3) | 12.5           | 12.0 |
| 13 | -  | -  | 6    | M10 x 18(n,8) | 45° | 90°            | 175 | -   | -   | 152    | -   | -   | 14(n,4)  | -        | -       | 210 | -   | -   | 200 | -  | -   | 10(10) | 4              | -  | 38.3(41.3) | 12.5           | 16.0 |
| 15 | -  | -  | 6    | M10 x 18(n,8) | 45° | 45°            | 230 | -   | -   | 170    | -   | -   | 14(n,8)  | -        | -       | 280 | -   | -   | 260 | -  | -   | 12     | 5              | M6 | 45.3       | 16.0           | 39.2 |
| 15 | -  | -  | 6    | M12 x 21(n,8) | 45° | 22.5°          | 255 | -   | -   | 180    | -   | -   | 16(n,8)  | -        | -       | 320 | -   | -   | 290 | -  | -   | 14     | 6              | M6 | 48.8       | 21.5           | 55.0 |
| 15 | -  | -  | 6    | M12 x 21(n,8) | 45° | 22.5°          | 255 | -   | -   | 180    | -   | -   | 16(n,8)  | -        | -       | 320 | -   | -   | 290 | -  | -   | 14     | 6              | M6 | 53.8       | 21.5           | 93   |

## Double MCFN ( worm-gear ) series installation dimensions

### MCFNDK



### MCFNDKO



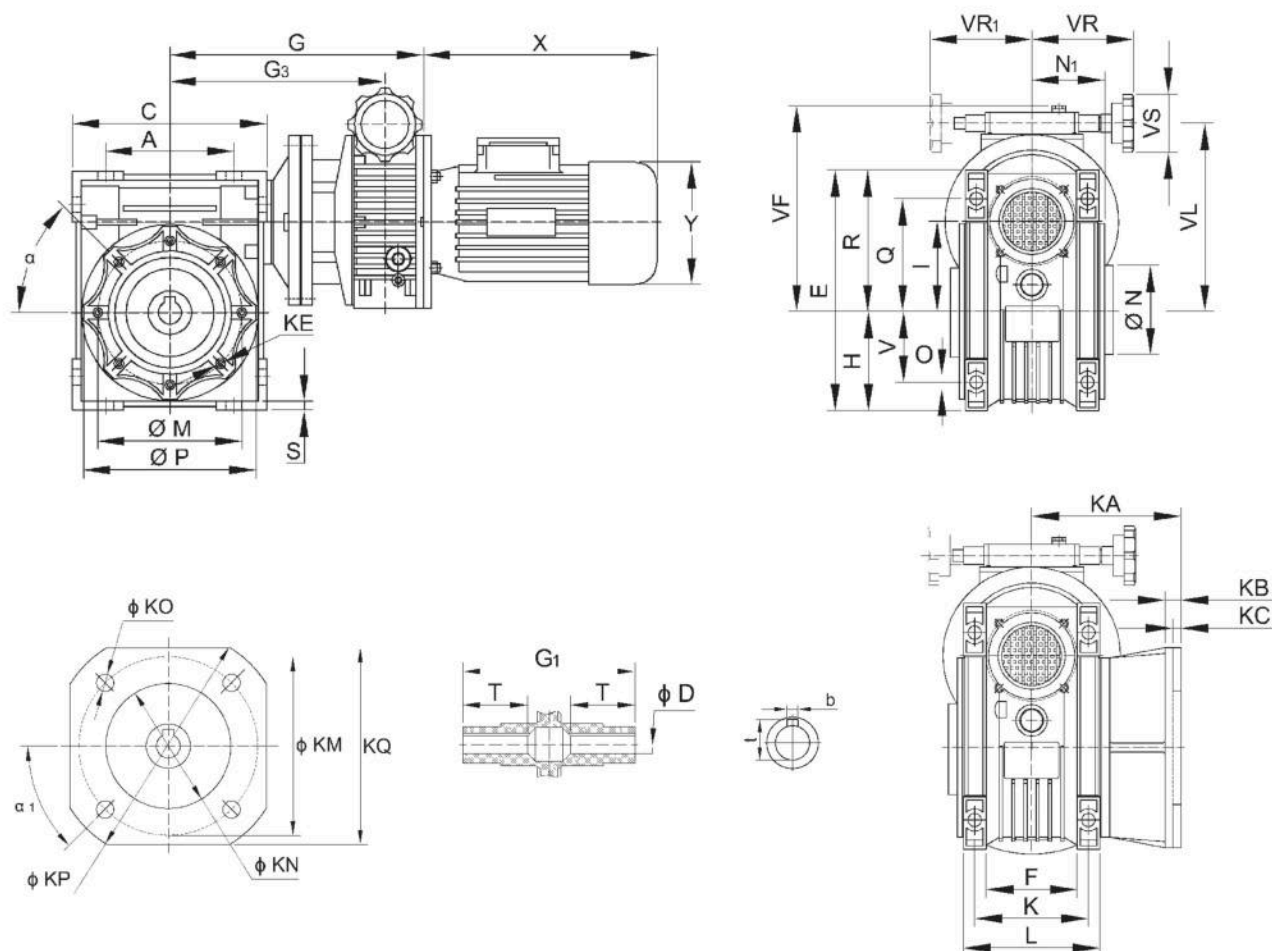
| Size       | A   | C     | D(H7)  | E     | F  | G   | G <sub>1</sub> | H     | I   | I1 | L   | M   | N(h8) | N <sub>1</sub> | O   | P   | Q   | R     | S   | T  | V   | K   | KA  |      |     | KB |    |    | KC   | KE            | α   | α <sub>1</sub> |
|------------|-----|-------|--------|-------|----|-----|----------------|-------|-----|----|-----|-----|-------|----------------|-----|-----|-----|-------|-----|----|-----|-----|-----|------|-----|----|----|----|------|---------------|-----|----------------|
|            |     |       |        |       |    |     |                |       |     |    |     |     |       |                |     |     |     |       |     |    |     |     | F   | FB   | FL  | F  | FB | FL |      |               |     |                |
| 40/63      | 70  | 100   | 18(19) | 121.5 | 43 | 123 | 78             | 50    | 40  | 40 | 71  | 75  | 60    | 36.5           | 6.5 | 87  | 55  | 71.5  | 6.5 | 26 | 35  | 60  | 67  | 76.5 | 97  | 7  | 9  | 7  | 4(5) | M6 x 8(n,4)   | 45° | 45°            |
| 50/63      | 80  | 120   | 25(24) | 144   | 49 | 133 | 92             | 60    | 50  | 40 | 85  | 85  | 70    | 43.5           | 8.5 | 100 | 64  | 84    | 7   | 30 | 40  | 70  | 90  | 87.5 | 120 | 9  | 10 | 9  | 5(5) | M8 x 10(n,4)  | 45° | 45°            |
| 50/71      | 80  | 120   | 25(24) | 144   | 49 | 143 | 92             | 60    | 50  | 50 | 85  | 85  | 70    | 43.5           | 8.5 | 100 | 64  | 84    | 7   | 30 | 40  | 70  | 90  | 87.5 | 120 | 9  | 10 | 9  | 5(5) | M8 x 10(n,4)  | 45° | 45°            |
| 63/63      | 100 | 144   | 25(28) | 174   | 67 | 148 | 112            | 72    | 63  | 40 | 103 | 95  | 80    | 53             | 8.5 | 110 | 80  | 102   | 8   | 36 | 50  | 85  | 82  | 99   | 112 | 10 | 11 | 10 | 6(5) | M8 x 14(n,8)  | 45° | 45°            |
| 63/71      | 100 | 144   | 25(28) | 174   | 67 | 158 | 112            | 72    | 63  | 50 | 103 | 95  | 80    | 53             | 8.5 | 110 | 80  | 102   | 8   | 36 | 50  | 85  | 82  | 99   | 112 | 10 | 11 | 10 | 6(5) | M8 x 14(n,8)  | 45° | 45°            |
| 75/71      | 120 | 172   | 28(35) | 205   | 72 | 176 | 120            | 86    | 75  | 50 | 112 | 115 | 95    | 57             | 11  | 140 | 93  | 119   | 10  | 40 | 60  | 90  | 111 | -    | -   | 13 | -  | -  | 6    | M8 x 14(n,8)  | 45° | 45°            |
| 75/80      | 120 | 172   | 28(35) | 205   | 72 | 186 | 120            | 86    | 75  | 63 | 112 | 115 | 95    | 57             | 11  | 140 | 93  | 119   | 10  | 40 | 60  | 90  | 111 | -    | -   | 13 | -  | -  | 6    | M8 x 14(n,8)  | 45° | 45°            |
| 90/71      | 140 | 208   | 35(38) | 238   | 74 | 193 | 140            | 103   | 90  | 50 | 130 | 130 | 110   | 67             | 13  | 160 | 102 | 135   | 11  | 45 | 70  | 100 | 111 | -    | -   | 13 | -  | -  | 6    | M10 x 18(n,8) | 45° | 45°            |
| 90/80      | 140 | 208   | 35(38) | 238   | 74 | 203 | 140            | 103   | 90  | 63 | 130 | 130 | 110   | 67             | 13  | 160 | 102 | 135   | 11  | 45 | 70  | 100 | 111 | -    | -   | 13 | -  | -  | 6    | M10 x 18(n,8) | 45° | 45°            |
| 110/80(90) | 170 | 252.5 | 42     | 295   | -  | 233 | 155            | 127.5 | 110 | 63 | 144 | 165 | 130   | 74             | 14  | 200 | 125 | 167.5 | 14  | 50 | 85  | 115 | 131 | -    | -   | 15 | -  | -  | 6    | M10 x 18(n,8) | 45° | 45°            |
| 130/80(90) | 200 | 292.5 | 45     | 335   | -  | 253 | 170            | 147.5 | 130 | 63 | 155 | 215 | 180   | 81             | 16  | 250 | 140 | 187.5 | 15  | 60 | 100 | 120 | 140 | -    | -   | 15 | -  | -  | 6    | M12 x 21(n,8) | 45° | 45°            |

| Size       | KM  |     |     | KN(H8) |     |     | KO      |          |         | KP  |     |     | KQ  |    |     | b      | t          | Kg   |
|------------|-----|-----|-----|--------|-----|-----|---------|----------|---------|-----|-----|-----|-----|----|-----|--------|------------|------|
|            | F   | FB  | FL  | F      | FB  | FL  | F       | FB       | FL      | F   | FB  | FL  | F   | FB | FL  |        |            |      |
| 40/63      | 87  | 115 | 87  | 60     | 95  | 60  | 9(n,4)  | 9.5(n,4) | 9(n,4)  | 110 | 140 | 110 | 95  | -  | 95  | 6(6)   | 20.8(21.8) | 3.9  |
| 50/63      | 90  | 130 | 90  | 70     | 110 | 70  | 11(n,4) | 9.5(n,4) | 11(n,4) | 125 | 160 | 125 | 110 | -  | 110 | 8(8)   | 28.3(27.3) | 5.2  |
| 50/71      | 90  | 130 | 90  | 70     | 110 | 70  | 11(n,4) | 9.5(n,4) | 11(n,4) | 125 | 160 | 125 | 110 | -  | 110 | 8(8)   | 28.3(27.3) | 5.8  |
| 63/63      | 150 | 165 | 150 | 115    | 130 | 115 | 11(n,4) | 11(n,4)  | 11(n,4) | 180 | 200 | 180 | 142 | -  | 142 | 8(8)   | 28.3(31.3) | 7.9  |
| 63/71      | 150 | 165 | 150 | 115    | 130 | 115 | 11(n,4) | 11(n,4)  | 11(n,4) | 180 | 200 | 180 | 142 | -  | 142 | 8(8)   | 28.3(31.3) | 8.5  |
| 75/71      | 165 | -   | -   | 130    | -   | -   | 14(n,4) | -        | -       | 200 | -   | -   | 170 | -  | -   | 8(10)  | 31.3(38.3) | 11.3 |
| 75/80      | 165 | -   | -   | 130    | -   | -   | 14(n,4) | -        | -       | 200 | -   | -   | 170 | -  | -   | 8(10)  | 31.3(38.3) | 13.1 |
| 90/71      | 175 | -   | -   | 152    | -   | -   | 14(n,4) | -        | -       | 210 | -   | -   | 200 | -  | -   | 10(10) | 38.3(41.3) | 15.3 |
| 90/80      | 175 | -   | -   | 152    | -   | -   | 14(n,4) | -        | -       | 210 | -   | -   | 200 | -  | -   | 10(10) | 38.3(41.3) | 17.2 |
| 110/80(90) | 230 | -   | -   | 170    | -   | -   | 14(n,8) | -        | -       | 280 | -   | -   | 260 | -  | -   | 12     | 45.3       | 39   |
| 130/80(90) | 235 | -   | -   | 180    | -   | -   | 16(n,8) | -        | -       | 320 | -   | -   | 290 | -  | -   | 14     | 48.8       | 52.2 |



## MCNUD ( worm-variator ) series installation dimensions

### MCNUD



| Size    | a   | a1    | A   | K   | KC |    |    | KE            | KM  |     |     | KN(H8) |     |     | KO      |          |         | M   | N(h8) | N1   | O   | Q   | S   | V   | b      | D(H7)  | t          | T  |
|---------|-----|-------|-----|-----|----|----|----|---------------|-----|-----|-----|--------|-----|-----|---------|----------|---------|-----|-------|------|-----|-----|-----|-----|--------|--------|------------|----|
|         |     |       |     |     | F  | FB | FL |               | F   | FB  | FL  | F      | FB  | FL  | F       | FB       | FL      |     |       |      |     |     |     |     |        |        |            |    |
| 40/0.18 | 45° | 45°   | 70  | 60  | 4  | 5  | 4  | M6 x 8(n,4)   | 87  | 115 | 87  | 60     | 95  | 60  | 9(n,4)  | 9.5(n,4) | 9(n,4)  | 75  | 60    | 36.5 | 6.5 | 55  | 6.5 | 35  | 6(6)   | 18(19) | 20.8(21.8) | 26 |
| 50/0.18 | 45° | 45°   | 80  | 70  | 5  | 5  | 5  | M8 x 10(n,4)  | 90  | 130 | 90  | 70     | 110 | 70  | 11(n,4) | 9.5(n,4) | 11(n,4) | 85  | 70    | 43.5 | 8.5 | 64  | 7   | 40  | 8(8)   | 25(24) | 28.3(27.3) | 30 |
| 50/0.37 |     |       |     |     |    |    |    |               |     |     |     |        |     |     |         |          |         |     |       |      |     |     |     |     |        |        |            |    |
| 63/0.37 | 45° | 45°   | 100 | 85  | 6  | 5  | 6  | M8 x 14(n,8)  | 150 | 165 | 150 | 115    | 130 | 115 | 11(n,4) | 11(n,4)  | 11(n,4) | 95  | 80    | 53   | 8.5 | 80  | 8   | 50  | 8(8)   | 25(28) | 28.3(31.3) | 36 |
| 63/0.55 |     |       |     |     |    |    |    |               |     |     |     |        |     |     |         |          |         |     |       |      |     |     |     |     |        |        |            |    |
| 63/0.75 |     |       |     |     |    |    |    |               |     |     |     |        |     |     |         |          |         |     |       |      |     |     |     |     |        |        |            |    |
| 75/0.37 |     |       |     |     |    |    |    |               |     |     |     |        |     |     |         |          |         |     |       |      |     |     |     |     |        |        |            |    |
| 75/0.55 | 45° | 45°   | 120 | 90  | 6  | -  | -  | M8 x 14(n,8)  | 165 | -   | -   | 130    | -   | -   | 14(n,4) | -        | -       | 115 | 95    | 57   | 11  | 93  | 10  | 60  | 8(10)  | 28(35) | 31.3(38.3) | 40 |
| 75/0.75 |     |       |     |     |    |    |    |               |     |     |     |        |     |     |         |          |         |     |       |      |     |     |     |     |        |        |            |    |
| 75/1.1  |     |       |     |     |    |    |    |               |     |     |     |        |     |     |         |          |         |     |       |      |     |     |     |     |        |        |            |    |
| 75/1.5  |     |       |     |     |    |    |    |               |     |     |     |        |     |     |         |          |         |     |       |      |     |     |     |     |        |        |            |    |
| 90/0.55 |     |       |     |     |    |    |    |               |     |     |     |        |     |     |         |          |         |     |       |      |     |     |     |     |        |        |            |    |
| 90/0.75 | 45° | 45°   | 140 | 100 | 6  | -  | -  | M10 x 18(n,8) | 175 | -   | -   | 152    | -   | -   | 14(n,4) | -        | -       | 130 | 110   | 67   | 13  | 102 | 11  | 70  | 10(10) | 35(38) | 38.3(41.3) | 45 |
| 90/1.1  |     |       |     |     |    |    |    |               |     |     |     |        |     |     |         |          |         |     |       |      |     |     |     |     |        |        |            |    |
| 90/1.5  |     |       |     |     |    |    |    |               |     |     |     |        |     |     |         |          |         |     |       |      |     |     |     |     |        |        |            |    |
| 110/1.1 |     |       |     |     |    |    |    |               |     |     |     |        |     |     |         |          |         |     |       |      |     |     |     |     |        |        |            |    |
| 110/1.5 |     |       |     |     |    |    |    |               |     |     |     |        |     |     |         |          |         |     |       |      |     |     |     |     |        |        |            |    |
| 110/2.2 | 45° | 45°   | 170 | 115 | 6  | -  | -  | M10 x 18(n,8) | 230 | -   | -   | 170    | -   | -   | 14(n,8) | -        | -       | 165 | 130   | 74   | 14  | 125 | 14  | 85  | 12     | 42     | 45.3       | 50 |
| 110/3.0 |     |       |     |     |    |    |    |               |     |     |     |        |     |     |         |          |         |     |       |      |     |     |     |     |        |        |            |    |
| 110/4.0 |     |       |     |     |    |    |    |               |     |     |     |        |     |     |         |          |         |     |       |      |     |     |     |     |        |        |            |    |
| 130/1.5 |     |       |     |     |    |    |    |               |     |     |     |        |     |     |         |          |         |     |       |      |     |     |     |     |        |        |            |    |
| 130/2.2 | 45° | 22.5° | 200 | 120 | 6  | -  | -  | M12 x 21(n,8) | 255 | -   | -   | 180    | -   | -   | 16(n,8) | -        | -       | 215 | 180   | 81   | 16  | 140 | 15  | 100 | 14     | 45     | 48.8       | 60 |
| 130/3.0 |     |       |     |     |    |    |    |               |     |     |     |        |     |     |         |          |         |     |       |      |     |     |     |     |        |        |            |    |
| 130/4.0 |     |       |     |     |    |    |    |               |     |     |     |        |     |     |         |          |         |     |       |      |     |     |     |     |        |        |            |    |

## Continuos

| Size    | C     | E     | F  | G     | G <sub>1</sub> | G <sub>3</sub> | H   | I   | KA  |      |     | KB |    |    | KP  |     |     | KQ  |    |     | L   | P   | R    | VF  | VL  | VS  | VR  | VR <sub>1</sub> |
|---------|-------|-------|----|-------|----------------|----------------|-----|-----|-----|------|-----|----|----|----|-----|-----|-----|-----|----|-----|-----|-----|------|-----|-----|-----|-----|-----------------|
|         |       |       |    |       |                |                |     |     | F   | FB   | FL  | F  | FB | FL | F   | FB  | FL  | F   | FB | FL  |     |     |      |     |     |     |     |                 |
| 40/0.18 | 100   | 121.5 | 43 | 183   | 78             | 134            | 50  | 40  | 67  | 76.5 | 97  | 7  | 9  | 7  | 110 | 140 | 110 | 95  | -  | 95  | 71  | 87  | 71.5 | 151 | 118 | 85  | 110 | 110             |
| 50/0.18 | 120   | 144   | 49 | 193   | 92             | 145            | 60  | 50  | 90  | 87.5 | 120 | 9  | 10 | 9  | 125 | 160 | 125 | 110 | -  | 110 | 85  | 100 | 84   | 161 | 128 | 85  | 110 | 110             |
| 50/0.37 |       |       |    | 190   |                | 154            |     |     |     |      |     |    |    |    |     |     |     |     |    |     |     |     |      | 173 | 140 | 85  | 110 | 110             |
| 63/0.37 | 144   | 174   | 67 | 205   | 112            | 169            | 72  | 63  | 82  | 99   | 112 | 10 | 11 | 10 | 180 | 200 | 180 | 142 | -  | 142 | 103 | 110 | 102  | 186 | 153 | 85  | 110 | 110             |
| 63/0.55 |       |       |    | 234   |                | 181            |     |     |     |      |     |    |    |    |     |     |     |     |    |     |     |     |      | 203 | 170 | 110 | 120 | 120             |
| 63/0.75 |       |       |    | 234   |                | 181            |     |     |     |      |     |    |    |    |     |     |     |     |    |     |     |     |      | 203 | 170 | 110 | 120 | 120             |
| 75/0.37 | 172   | 205   | 72 | 223   | 120            | 187            | 86  | 75  | 111 | -    | -   | 13 | -  | -  | 200 | -   | -   | 170 | -  | -   | 112 | 140 | 119  | 198 | 165 | 85  | 110 | 110             |
| 75/0.55 |       |       |    | 252   |                | 198            |     |     |     |      |     |    |    |    |     |     |     |     |    |     |     |     |      | 215 | 182 | 110 | 120 | 120             |
| 75/0.75 |       |       |    | 252   |                | 198            |     |     |     |      |     |    |    |    |     |     |     |     |    |     |     |     |      | 215 | 182 | 110 | 120 | 120             |
| 75/1.1  |       |       |    | 259.5 |                | 207.5          |     |     |     |      |     |    |    |    |     |     |     |     |    |     |     |     |      | 199 | 177 | 110 | 150 | -               |
| 75/1.5  |       |       |    | 300.5 |                | 227.5          |     |     |     |      |     |    |    |    |     |     |     |     |    |     |     |     |      | 219 | 197 | 110 | 150 | -               |
| 90/0.55 | 208   | 238   | 74 | 269   | 140            | 215            | 103 | 90  | 111 | -    | -   | 13 | -  | -  | 210 | -   | -   | 200 | -  | -   | 130 | 160 | 135  | 230 | 197 | 110 | 120 | 120             |
| 90/0.75 |       |       |    | 269   |                | 215            |     |     |     |      |     |    |    |    |     |     |     |     |    |     |     |     |      | 230 | 197 | 110 | 120 | 120             |
| 90/1.1  |       |       |    | 276.5 |                | 224.5          |     |     |     |      |     |    |    |    |     |     |     |     |    |     |     |     |      | 214 | 192 | 110 | 150 | -               |
| 90/1.5  |       |       |    | 317.5 |                | 244.5          |     |     |     |      |     |    |    |    |     |     |     |     |    |     |     |     |      | 234 | 212 | 110 | 150 | -               |
| 110/1.1 | 252.5 | 295   | -  | 307   | 155            | 255            | 128 | 110 | 131 | -    | -   | 15 | -  | -  | 280 | -   | -   | 260 | -  | -   | 144 | 200 | 168  | 234 | 212 | 110 | 120 | -               |
| 110/1.5 |       |       |    | 348   |                | 275            |     |     |     |      |     |    |    |    |     |     |     |     |    |     |     |     |      | 254 | 232 | 110 | 150 | -               |
| 110/2.2 |       |       |    | 368   |                | 291            |     |     |     |      |     |    |    |    |     |     |     |     |    |     |     |     |      | 298 | 260 | 110 | 160 | -               |
| 110/3.0 |       |       |    | 368   |                | 291            |     |     |     |      |     |    |    |    |     |     |     |     |    |     |     |     |      | 298 | 260 | 110 | 160 | -               |
| 110/4.0 | 292.5 | 335   | -  | 368   | 170            | 291            | 148 | 130 | 140 | -    | -   | 15 | -  | -  | 320 | -   | -   | 290 | -  | -   | 155 | 250 | 188  | 298 | 260 | 110 | 160 | -               |
| 130/1.5 |       |       |    | 368   |                | 295            |     |     |     |      |     |    |    |    |     |     |     |     |    |     |     |     |      | 274 | 252 | 110 | 150 | -               |
| 130/2.2 |       |       |    | 388   |                | 311            |     |     |     |      |     |    |    |    |     |     |     |     |    |     |     |     |      | 318 | 280 | 110 | 160 | -               |
| 130/3.0 |       |       |    | 388   |                | 311            |     |     |     |      |     |    |    |    |     |     |     |     |    |     |     |     |      | 318 | 280 | 110 | 160 | -               |
| 130/4.0 |       |       |    | 388   |                | 311            |     |     |     |      |     |    |    |    |     |     |     |     |    |     |     |     |      | 318 |     | 110 | 160 | -               |

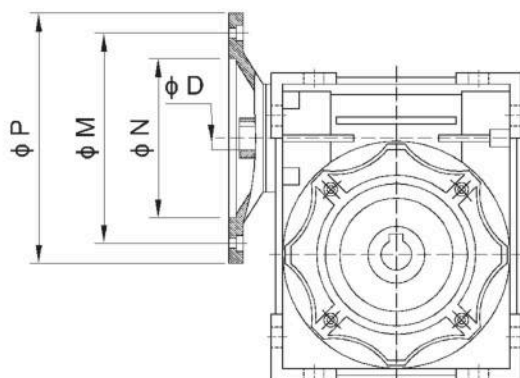
## Motor size

| Frame No.    | 63   | 71   | 80   | 90S | 90L | 100L | 100L | 112M |
|--------------|------|------|------|-----|-----|------|------|------|
| 4P /KW Power | 0.18 | 0.37 | 0.75 | 1.1 | 1.5 | 2.2  | 3.0  | 4.0  |
| X            | 200  | 227  | 268  | 265 | 290 | 320  | 320  | 340  |
| Y            | 120  | 141  | 160  | 195 | 195 | 215  | 215  | 240  |

## Accessories dimensions

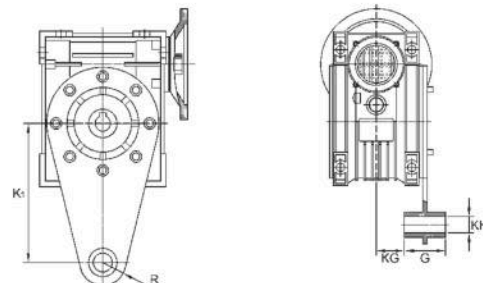
( D )

### Motor Mounting Facility



| Size | PAM<br>IEC | N   | M   | P   | 7.5 | 10 | 15 | 20 | 25 | 30 | 40 | 50 | 60 | 80 | 100 |
|------|------------|-----|-----|-----|-----|----|----|----|----|----|----|----|----|----|-----|
| 25   | 56B14      | 50  | 65  | 80  | 9   | 9  | 9  | 9  | -  | 9  | 9  | 9  | 9  | -  | -   |
| 30   | 63B5       | 95  | 115 | 140 | 11  | 11 | 11 | 11 | 11 | 11 | 11 | 11 | -  | -  | -   |
|      | 63B14      | 60  | 75  | 90  | -   | -  | -  | -  | -  | -  | -  | -  | -  | -  | -   |
|      | 56B5       | 80  | 100 | 120 | 9   | 9  | 9  | 9  | 9  | 9  | 9  | 9  | 9  | 9  | -   |
|      | 56B14      | 50  | 65  | 80  | -   | -  | -  | -  | -  | -  | -  | -  | -  | -  | -   |
| 40   | 71B5       | 110 | 130 | 160 | 14  | 14 | 14 | 14 | 14 | 14 | 14 | -  | -  | -  | -   |
|      | 71B14      | 70  | 85  | 105 | -   | -  | -  | -  | -  | -  | -  | -  | -  | -  | -   |
|      | 63B5       | 95  | 115 | 140 | 11  | 11 | 11 | 11 | 11 | 11 | 11 | 11 | 11 | 11 | 11  |
|      | 63B14      | 60  | 75  | 90  | -   | -  | -  | -  | -  | -  | -  | -  | -  | -  | -   |
| 50   | 56B5       | 80  | 100 | 120 | -   | -  | -  | -  | -  | -  | -  | 9  | 9  | 9  | 9   |
|      | 80B5       | 130 | 165 | 200 | 19  | 19 | 19 | 19 | 19 | 19 | -  | -  | -  | -  | -   |
|      | 80B14      | 80  | 100 | 120 | -   | -  | -  | -  | -  | -  | -  | -  | -  | -  | -   |
|      | 71B5       | 110 | 130 | 160 | 14  | 14 | 14 | 14 | 14 | 14 | 14 | 14 | 14 | 14 | -   |
| 63   | 63B5       | 95  | 115 | 140 | -   | -  | -  | -  | -  | -  | 11 | 11 | 11 | 11 | 11  |
|      | 90B5       | 130 | 165 | 200 | 24  | 24 | 24 | 24 | 24 | 24 | -  | -  | -  | -  | -   |
|      | 90B14      | 95  | 115 | 140 | -   | -  | -  | -  | -  | -  | -  | -  | -  | -  | -   |
|      | 80B5       | 130 | 165 | 200 | 19  | 19 | 19 | 19 | 19 | 19 | 19 | 19 | 19 | 19 | -   |
| 75   | 80B14      | 80  | 100 | 120 | -   | -  | -  | -  | -  | -  | -  | -  | -  | -  | -   |
|      | 71B5       | 110 | 130 | 160 | -   | -  | -  | -  | -  | -  | 14 | 14 | 14 | 14 | 14  |
|      | 100/112B5  | 180 | 215 | 250 | 28  | 28 | 28 | -  | -  | -  | -  | -  | -  | -  | -   |
|      | 100/112B14 | 110 | 130 | 160 | -   | -  | -  | -  | -  | -  | -  | -  | -  | -  | -   |
| 90   | 90B5       | 130 | 165 | 200 | 24  | 24 | 24 | 24 | 24 | 24 | 24 | -  | -  | -  | -   |
|      | 90B14      | 95  | 115 | 140 | -   | -  | -  | 19 | 19 | 19 | 19 | 19 | 19 | 19 | 19  |
|      | 80B5       | 130 | 165 | 200 | -   | -  | -  | -  | -  | -  | -  | 19 | 19 | 19 | 19  |
|      | 80B14      | 80  | 100 | 120 | -   | -  | -  | -  | -  | -  | -  | 14 | 14 | 14 | 14  |
| 110  | 100/112B5  | 180 | 215 | 250 | 28  | 28 | 28 | 28 | 28 | 28 | -  | -  | -  | -  | -   |
|      | 100/112B14 | 110 | 130 | 160 | -   | -  | -  | -  | -  | -  | -  | -  | -  | -  | -   |
|      | 90B5       | 130 | 165 | 200 | 24  | 24 | 24 | 24 | 24 | 24 | 24 | 24 | 24 | 24 | 24  |
|      | 90B14      | 95  | 115 | 140 | -   | -  | -  | -  | -  | -  | -  | -  | -  | -  | -   |
| 130  | 80B5       | 130 | 165 | 200 | -   | -  | -  | -  | -  | -  | 19 | 19 | 19 | 19 | 19  |
|      | 132B5      | 230 | 265 | 300 | 38  | 38 | 38 | 38 | 38 | 38 | 38 | -  | -  | -  | -   |
|      | 100/112B5  | 180 | 215 | 250 | -   | -  | -  | -  | -  | -  | 28 | 28 | 28 | 28 | 28  |
|      | 90B5       | 130 | 165 | 200 | -   | -  | -  | -  | -  | -  | -  | -  | -  | 24 | 24  |
| 150  | 160B5      | 250 | 300 | 350 | 42  | 42 | 42 | 42 | -  | -  | -  | -  | -  | -  | -   |
|      | 132B5      | 230 | 265 | 300 | -   | -  | -  | 38 | 38 | 38 | 38 | 38 | -  | -  | -   |
|      | 100/112B5  | 180 | 215 | 250 | -   | -  | -  | -  | -  | -  | -  | 28 | 28 | 28 | 28  |

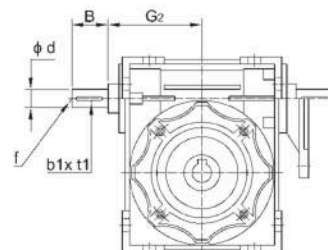
### Torque Arm



| Size | K <sub>1</sub> | G  | KG   | KH | R  |
|------|----------------|----|------|----|----|
| 25   | 70             | 14 | 17.5 | 8  | 15 |
| 30   | 85             | 14 | 24   | 8  | 15 |
| 40   | 100            | 14 | 31.5 | 10 | 18 |
| 50   | 100            | 14 | 38.5 | 10 | 18 |
| 63   | 150            | 14 | 49   | 10 | 18 |
| 75   | 200            | 25 | 47.5 | 20 | 30 |
| 90   | 200            | 25 | 57.5 | 20 | 30 |
| 110  | 250            | 30 | 62   | 25 | 35 |
| 130  | 250            | 30 | 69   | 25 | 35 |
| 150  | 250            | 30 | 84   | 25 | 35 |

( E )

### Double Worm Shaft



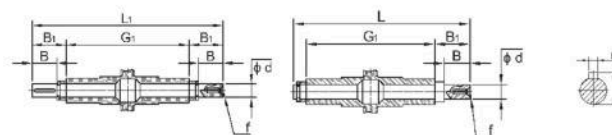
| Size | G <sub>2</sub> | d (j6) | B  | f   | b <sub>1</sub> | t <sub>1</sub> |
|------|----------------|--------|----|-----|----------------|----------------|
| 30   | 45             | 9      | 20 | -   | 3              | 10.2           |
| 40   | 53             | 11     | 23 | -   | 4              | 12.5           |
| 50   | 64             | 14     | 30 | M6  | 5              | 16             |
| 63   | 75             | 19     | 40 | M6  | 6              | 21.5           |
| 75   | 90             | 24     | 50 | M8  | 8              | 27             |
| 90   | 108            | 24     | 50 | M8  | 8              | 27             |
| 110  | 135            | 28     | 60 | M10 | 8              | 31             |
| 130  | 155            | 30     | 80 | M10 | 8              | 33             |
| 150  | 175            | 35     | 80 | M12 | 10             | 38             |

( DZ ), ( SZ )

### Sizes of Single(DZ) & Double(SZ) Output shaft

SZ

DZ



| Size | d (h6) | B  | B <sub>1</sub> | G <sub>1</sub> | L   | L <sub>1</sub> | f   | b <sub>1</sub> | t <sub>1</sub> |
|------|--------|----|----------------|----------------|-----|----------------|-----|----------------|----------------|
| 25   | 11     | 23 | 25.5           | 50             | 81  | 101            | -   | 4              | 12.5           |
| 30   | 14     | 30 | 32.5           | 63             | 102 | 128            | M6  | 5              | 16             |
| 40   | 18     | 40 | 43             | 78             | 128 | 164            | M6  | 6              | 20.5           |
| 50   | 25     | 50 | 53.5           | 92             | 153 | 199            | M10 | 8              | 28             |
| 63   | 25     | 50 | 53.5           | 112            | 173 | 219            | M10 | 8              | 28             |
| 75   | 28     | 60 | 63.5           | 120            | 192 | 247            | M10 | 8              | 31             |
| 90   | 35     | 80 | 84.5           | 140            | 234 | 309            | M12 | 10             | 38             |
| 110  | 42     | 80 | 84.5           | 155            | 249 | 324            | M16 | 12             | 45             |
| 130  | 45     | 80 | 85             | 170            | 265 | 340            | M16 | 14             | 48.5           |
| 150  | 50     | 82 | 87             | 200            | 297 | 374            | M16 | 14             | 53.5           |



## MCN

### Performance parameter of Single MCN series

| Input                                      | Size | i   | n <sub>2</sub> (r/min) | M <sub>2</sub> (N·m) |
|--------------------------------------------|------|-----|------------------------|----------------------|
| 0.06kw<br>4P<br>n <sub>1</sub> = 1400r/min | 25   | 7.5 | 186.7                  | 2.6                  |
|                                            | 30   | 7.5 | 186.7                  | 2.6                  |
|                                            | 25   | 10  | 140                    | 3.4                  |
|                                            | 30   | 10  | 140                    | 3.4                  |
|                                            | 25   | 15  | 93.3                   | 4.9                  |
|                                            | 30   | 15  | 93.3                   | 4.7                  |
|                                            | 25   | 20  | 70                     | 6.1                  |
|                                            | 30   | 20  | 70                     | 6                    |
|                                            | 30   | 25  | 56                     | 7                    |
|                                            | 25   | 30  | 46.7                   | 8.2                  |
|                                            | 30   | 30  | 46.7                   | 8                    |
|                                            | 25   | 40  | 35                     | 10.2                 |
|                                            | 30   | 40  | 35                     | 9.7                  |
|                                            | 25   | 50  | 28                     | 11                   |
|                                            | 30   | 50  | 28                     | 11.3                 |
|                                            | 40   | 50  | 28                     | 12.7                 |
|                                            | 25   | 60  | 23.3                   | 11                   |
|                                            | 30   | 60  | 23.3                   | 12.5                 |
|                                            | 40   | 60  | 23.3                   | 14.2                 |
|                                            | 30   | 80  | 17.5                   | 12.5                 |
| 0.09kw<br>4P<br>n <sub>1</sub> = 1400r/min | 40   | 80  | 17.5                   | 17                   |
|                                            | 40   | 100 | 14                     | 19.2                 |
|                                            | 25   | 7.5 | 186.7                  | 3.9                  |
|                                            | 30   | 7.5 | 186.7                  | 3.9                  |
|                                            | 25   | 10  | 140                    | 5.1                  |
|                                            | 30   | 10  | 140                    | 5                    |
|                                            | 25   | 15  | 93.3                   | 7.3                  |
|                                            | 30   | 15  | 93.3                   | 7.1                  |
|                                            | 25   | 20  | 70                     | 9.2                  |
|                                            | 30   | 20  | 70                     | 9                    |
|                                            | 30   | 25  | 56                     | 10.4                 |
|                                            | 25   | 30  | 46.7                   | 12.3                 |
|                                            | 30   | 30  | 46.7                   | 12                   |
|                                            | 25   | 40  | 35                     | 13                   |
|                                            | 30   | 40  | 35                     | 14.5                 |
|                                            | 30   | 50  | 28                     | 16.9                 |
|                                            | 40   | 50  | 28                     | 19                   |
|                                            | 30   | 60  | 23.3                   | 16.9                 |
|                                            | 40   | 60  | 23.3                   | 21.4                 |
|                                            | 40   | 80  | 17.5                   | 25.5                 |
| 0.12kw<br>4P<br>n <sub>1</sub> = 1400r/min | 40   | 100 | 14                     | 28.9                 |
|                                            | 30   | 7.5 | 186.7                  | 5.2                  |
|                                            | 40   | 7.5 | 186.7                  | 5.3                  |
|                                            | 30   | 10  | 140                    | 6.7                  |
|                                            | 40   | 10  | 140                    | 7                    |
|                                            | 30   | 15  | 93.3                   | 9.5                  |
|                                            | 40   | 15  | 93.3                   | 10.1                 |
|                                            | 30   | 20  | 70                     | 12                   |
|                                            | 40   | 20  | 70                     | 12.8                 |
|                                            | 30   | 25  | 56                     | 13.9                 |
|                                            | 40   | 25  | 56                     | 15.3                 |
|                                            | 30   | 30  | 46.7                   | 16                   |
|                                            | 40   | 30  | 46.7                   | 17.2                 |
|                                            | 30   | 40  | 35                     | 17                   |
|                                            | 40   | 40  | 35                     | 21.3                 |
|                                            | 50   | 40  | 35                     | 21.9                 |
|                                            | 40   | 50  | 28                     | 25.4                 |
|                                            | 50   | 50  | 28                     | 25.8                 |
|                                            | 40   | 60  | 28.5                   | 28.5                 |

| Input                                      | Size | i   | n <sub>2</sub> (r/min) | M <sub>2</sub> (N·m) |
|--------------------------------------------|------|-----|------------------------|----------------------|
| 0.12kw<br>4P<br>n <sub>1</sub> = 1400r/min | 50   | 60  | 23.3                   | 29                   |
|                                            | 40   | 80  | 17.5                   | 34.1                 |
|                                            | 50   | 80  | 17.5                   | 34.7                 |
|                                            | 50   | 100 | 14                     | 40.1                 |
|                                            | 30   | 7.5 | 186.7                  | 8                    |
|                                            | 40   | 7.5 | 186.7                  | 8                    |
|                                            | 30   | 10  | 140                    | 10                   |
|                                            | 40   | 10  | 140                    | 10                   |
|                                            | 30   | 15  | 93.3                   | 14                   |
|                                            | 40   | 15  | 93.3                   | 15                   |
| 0.18kw<br>4P<br>n <sub>1</sub> = 1400r/min | 30   | 20  | 70                     | 18                   |
|                                            | 40   | 20  | 70                     | 19                   |
|                                            | 30   | 25  | 56                     | 20                   |
|                                            | 40   | 25  | 56                     | 23                   |
|                                            | 40   | 30  | 46.7                   | 26                   |
|                                            | 50   | 40  | 35                     | 32                   |
|                                            | 40   | 40  | 35                     | 32                   |
|                                            | 40   | 50  | 28                     | 38                   |
|                                            | 50   | 50  | 28                     | 38                   |
|                                            | 50   | 60  | 23.3                   | 34                   |
| 0.25kw<br>4P<br>n <sub>1</sub> = 1400r/min | 50   | 80  | 17.5                   | 53                   |
|                                            | 50   | 100 | 14                     | 55                   |
|                                            | 40   | 7.5 | 186.7                  | 11                   |
|                                            | 50   | 7.5 | 186.7                  | 11                   |
|                                            | 40   | 10  | 140                    | 14                   |
|                                            | 50   | 10  | 140                    | 14                   |
|                                            | 40   | 15  | 93.3                   | 20                   |
|                                            | 50   | 15  | 93.3                   | 21                   |
|                                            | 40   | 20  | 70                     | 26                   |
|                                            | 50   | 20  | 70                     | 26                   |
| 0.37kw<br>4P<br>n <sub>1</sub> = 1400r/min | 40   | 25  | 56                     | 31                   |
|                                            | 50   | 25  | 56                     | 32                   |
|                                            | 40   | 30  | 46.7                   | 36                   |
|                                            | 50   | 30  | 46.7                   | 36                   |
|                                            | 40   | 40  | 35                     | 44                   |
|                                            | 50   | 40  | 35                     | 45                   |
|                                            | 50   | 50  | 28                     | 53                   |
|                                            | 50   | 60  | 23.3                   | 60                   |
|                                            | 50   | 80  | 17.5                   | 65                   |
|                                            | 63   | 80  | 17.5                   | 77                   |
|                                            | 63   | 100 | 14                     | 85                   |
|                                            | 40   | 7.5 | 186.7                  | 16                   |
|                                            | 50   | 7.5 | 186.7                  | 16                   |
|                                            | 40   | 10  | 140                    | 21                   |
|                                            | 50   | 10  | 140                    | 21                   |
|                                            | 40   | 15  | 93.3                   | 30                   |
|                                            | 50   | 15  | 93.3                   | 31                   |
|                                            | 40   | 20  | 70                     | 39                   |
|                                            | 50   | 20  | 70                     | 39                   |
|                                            | 50   | 25  | 56                     | 47                   |
|                                            | 50   | 30  | 46.7                   | 54                   |
|                                            | 50   | 40  | 35                     | 66                   |
|                                            | 63   | 40  | 35                     | 70                   |
|                                            | 50   | 50  | 28                     | 73                   |
|                                            | 63   | 50  | 28                     | 83                   |
|                                            | 63   | 60  | 23.3                   | 95                   |
|                                            | 63   | 80  | 17.5                   | 114                  |
|                                            | 63   | 100 | 14                     | 118                  |



## MCN

### Performance parameter of Single MCN Series

| Input                              | Size | i   | $n_2$ (r/min) | $M_2$ (N·m) |
|------------------------------------|------|-----|---------------|-------------|
| 0.55kw<br>4P<br>$n_1 = 1400$ r/min | 40   | 7.5 | 186.7         | 24.5        |
|                                    | 50   | 7.5 | 186.7         | 25          |
|                                    | 40   | 10  | 140           | 32          |
|                                    | 50   | 10  | 140           | 32          |
|                                    | 50   | 15  | 93.3          | 46          |
|                                    | 63   | 15  | 93.3          | 46          |
|                                    | 50   | 20  | 70            | 59          |
|                                    | 63   | 20  | 70            | 60          |
|                                    | 63   | 25  | 56            | 72          |
|                                    | 63   | 30  | 46.7          | 80          |
|                                    | 63   | 40  | 35            | 104         |
|                                    | 75   | 40  | 35            | 108         |
|                                    | 63   | 50  | 28            | 123         |
|                                    | 75   | 50  | 28            | 129         |
|                                    | 75   | 60  | 23.3          | 146         |
|                                    | 75   | 80  | 17.5          | 180         |
|                                    | 75   | 100 | 14            | 180         |
| 0.75kw<br>4P<br>$n_1 = 1400$ r/min | 50   | 7.5 | 186.7         | 34          |
|                                    | 63   | 7.5 | 186.7         | 33          |
|                                    | 50   | 10  | 140           | 44          |
|                                    | 63   | 10  | 140           | 44          |
|                                    | 50   | 15  | 93.3          | 63          |
|                                    | 63   | 15  | 93.3          | 63          |
|                                    | 63   | 20  | 70            | 82          |
|                                    | 63   | 25  | 56            | 99          |
|                                    | 63   | 30  | 46.7          | 109         |
|                                    | 75   | 30  | 46.7          | 116         |
|                                    | 63   | 40  | 35            | 143         |
|                                    | 75   | 40  | 35            | 147         |
|                                    | 75   | 50  | 28            | 176         |
|                                    | 90   | 50  | 28            | 184         |
|                                    | 75   | 60  | 23.3          | 200         |
|                                    | 90   | 60  | 23.3          | 212         |
|                                    | 90   | 80  | 17.5          | 257         |
|                                    | 90   | 100 | 14            | 270         |
| 1.1kw<br>4P<br>$n_1 = 1400$ r/min  | 63   | 7.5 | 186.7         | 49          |
|                                    | 75   | 7.5 | 186.7         | 49          |
|                                    | 63   | 10  | 140           | 65          |
|                                    | 75   | 10  | 140           | 66          |
|                                    | 63   | 15  | 93.3          | 93          |
|                                    | 75   | 15  | 93.3          | 95          |
|                                    | 63   | 20  | 70            | 121         |
|                                    | 75   | 20  | 70            | 122         |
|                                    | 75   | 25  | 56            | 149         |
|                                    | 75   | 30  | 46.7          | 170         |
|                                    | 75   | 40  | 35            | 216         |
|                                    | 90   | 40  | 35            | 225         |
|                                    | 90   | 50  | 28            | 271         |
|                                    | 90   | 60  | 23.3          | 311         |
|                                    | 110  | 60  | 23.3          | 324         |
|                                    | 110  | 80  | 17.5          | 410         |
|                                    | 110  | 100 | 14            | 460         |
| 1.5kw<br>4P<br>$n_1 = 1400$ r/min  | 75   | 7.5 | 186.7         | 67          |
|                                    | 75   | 10  | 140           | 90          |
|                                    | 75   | 15  | 93.3          | 130         |
|                                    | 75   | 20  | 70            | 167         |
|                                    | 75   | 25  | 56            | 200         |
|                                    | 90   | 25  | 56            | 209         |
|                                    | 75   | 30  | 46.7          | 230         |
|                                    | 90   | 30  | 46.7          | 236         |
|                                    | 90   | 40  | 35            | 306         |
|                                    | 90   | 50  | 28            | 369         |
|                                    | 110  | 50  | 28            | 375         |
|                                    | 110  | 60  | 23.3          | 442         |
|                                    | 110  | 80  | 17.5          | 490         |
|                                    | 130  | 80  | 17.5          | 547         |
|                                    | 130  | 100 | 14            | 652         |
|                                    | 90   | 7.5 | 186.7         | 101         |
| 2.2kw<br>4P<br>$n_1 = 1400$ r/min  | 110  | 7.5 | 186.7         | 101         |
|                                    | 90   | 10  | 140           | 133         |
|                                    | 110  | 15  | 93.3          | 133         |

| Input                             | Size | i   | $n_2$ (r/min) | $M_2$ (N·m) |
|-----------------------------------|------|-----|---------------|-------------|
| 2.2kw<br>4P<br>$n_1 = 1400$ r/min | 90   | 15  | 93.3          | 193         |
|                                   | 110  | 15  | 93.3          | 193         |
|                                   | 90   | 20  | 70            | 251         |
|                                   | 110  | 20  | 70            | 256         |
|                                   | 90   | 25  | 56            | 307         |
|                                   | 110  | 25  | 56            | 316         |
|                                   | 90   | 30  | 46.7          | 346         |
|                                   | 110  | 30  | 46.7          | 355         |
|                                   | 110  | 40  | 35            | 462         |
|                                   | 110  | 50  | 28            | 550         |
|                                   | 130  | 50  | 28            | 566         |
|                                   | 150  | 50  | 28            | 570         |
|                                   | 130  | 60  | 23.3          | 650         |
|                                   | 150  | 60  | 23.3          | 657         |
|                                   | 130  | 80  | 17.5          | 803         |
|                                   | 150  | 80  | 17.5          | 816         |
|                                   | 150  | 100 | 14            | 960         |
| 3kw<br>4P<br>$n_1 = 1400$ r/min   | 110  | 7.5 | 186.7         | 138         |
|                                   | 110  | 10  | 140           | 182         |
|                                   | 110  | 15  | 93.3          | 263         |
|                                   | 110  | 20  | 70            | 348         |
|                                   | 110  | 25  | 56            | 430         |
|                                   | 110  | 30  | 46.7          | 484         |
|                                   | 110  | 40  | 35            | 631         |
|                                   | 130  | 40  | 35            | 638         |
|                                   | 130  | 50  | 28            | 767         |
|                                   | 150  | 50  | 28            | 778         |
|                                   | 130  | 60  | 23.3          | 884         |
|                                   | 150  | 60  | 23.3          | 896         |
|                                   | 150  | 80  | 17.5          | 1110        |
|                                   | 110  | 7.5 | 186.7         | 184         |
|                                   | 130  | 7.5 | 186.7         | 186         |
|                                   | 110  | 10  | 140           | 243         |
| 4kw<br>4P<br>$n_1 = 1400$ r/min   | 130  | 10  | 140           | 243         |
|                                   | 110  | 15  | 93.3          | 352         |
|                                   | 130  | 15  | 93.3          | 357         |
|                                   | 110  | 20  | 70            | 464         |
|                                   | 130  | 20  | 70            | 466         |
|                                   | 110  | 25  | 56            | 573         |
|                                   | 130  | 25  | 56            | 573         |
|                                   | 110  | 30  | 46.7          | 646         |
|                                   | 130  | 30  | 46.7          | 655         |
|                                   | 130  | 40  | 35            | 850         |
|                                   | 130  | 50  | 28            | 1023        |
|                                   | 150  | 50  | 28            | 1037        |
|                                   | 150  | 60  | 23.3          | 1195        |
|                                   | 110  | 7.5 | 186.7         | 253         |
|                                   | 130  | 7.5 | 186.7         | 256         |
|                                   | 110  | 10  | 140           | 334         |
| 5.5kw<br>4P<br>$n_1 = 1400$ r/min | 130  | 10  | 140           | 334         |
|                                   | 110  | 15  | 93.3          | 484         |
|                                   | 130  | 15  | 93.3          | 490         |
|                                   | 130  | 20  | 70            | 645         |
|                                   | 150  | 20  | 70            | 645         |
|                                   | 130  | 25  | 56            | 788         |
|                                   | 150  | 25  | 56            | 788         |
|                                   | 130  | 30  | 46.7          | 900         |
|                                   | 150  | 30  | 46.7          | 934         |
|                                   | 150  | 40  | 35            | 1171        |
|                                   | 110  | 7.5 | 186.7         | 345         |
|                                   | 130  | 7.5 | 186.7         | 349         |
|                                   | 110  | 10  | 140           | 455         |
|                                   | 130  | 10  | 140           | 455         |
|                                   | 130  | 15  | 93.3          | 667         |
|                                   | 150  | 20  | 70            | 880         |
|                                   | 150  | 25  | 56            | 1074        |
| 7.5kw<br>4P<br>$n_1 = 1400$ r/min | 150  | 7.5 | 186.7         | 512         |
|                                   | 150  | 10  | 140           | 675         |
|                                   | 150  | 15  | 93.3          | 990         |
| 11kw 4P<br>$n_1 = 1400$ r/min     | 150  | 7.5 | 186.7         | 698         |
|                                   | 150  | 10  | 140           | 921         |

## MCEN ( $n_1 = 1400\text{r/m}$ )

### Performance parameter of Double MCEN ( worm-worm ) Series

| Size  | i    | $n_2(\text{r/min})$ | $kW_1$ | $M_2(\text{N}\cdot\text{m})$ | $i_1$ | $i_2$ |
|-------|------|---------------------|--------|------------------------------|-------|-------|
| 25/30 | 100  | 14.0                | 0.09   | 30                           | 10    | 10    |
|       | 150  | 9.3                 | 0.06   | 28                           | 7.5   | 20    |
|       | 200  | 7.0                 | 0.06   | 28                           | 10    | 20    |
|       | 250  | 5.6                 | 0.06   | 35                           | 10    | 25    |
|       | 300  | 4.7                 | 0.06   | 31                           | 10    | 30    |
|       | 400  | 3.5                 | 0.06   | 28                           | 20    | 20    |
|       | 500  | 2.8                 | 0.06   | 34                           | 20    | 25    |
|       | 600  | 2.3                 | 0.06   | 31                           | 20    | 30    |
|       | 750  | 1.9                 | 0.06   | 34                           | 30    | 25    |
|       | 900  | 1.6                 | 0.06   | 31                           | 30    | 30    |
|       | 1200 | 1.2                 | 0.06   | 28                           | 30    | 40    |
|       | 1500 | 0.9                 | 0.06   | 26                           | 30    | 50    |
|       | 1800 | 0.8                 | 0.06   | 31                           | 60    | 30    |
|       | 2400 | 0.6                 | 0.06   | 28                           | 60    | 40    |
| 25/40 | 3000 | 0.5                 | 0.06   | 26                           | 60    | 50    |
|       | 300  | 4.7                 | 0.06   | 59                           | 10    | 30    |
|       | 400  | 3.5                 | 0.06   | 63                           | 10    | 40    |
|       | 500  | 2.8                 | 0.06   | 57                           | 10    | 50    |
|       | 600  | 2.3                 | 0.06   | 65                           | 15    | 40    |
|       | 750  | 1.9                 | 0.06   | 60                           | 15    | 50    |
|       | 900  | 1.6                 | 0.06   | 73                           | 30    | 30    |
|       | 1200 | 1.2                 | 0.06   | 65                           | 30    | 40    |
|       | 1500 | 0.9                 | 0.06   | 60                           | 30    | 50    |
|       | 1800 | 0.8                 | 0.06   | 56                           | 30    | 60    |
|       | 2400 | 0.6                 | 0.06   | 56                           | 40    | 60    |
|       | 3000 | 0.5                 | 0.06   | 60                           | 60    | 50    |
| 30/40 | 300  | 4.7                 | 0.09   | 70                           | 10    | 30    |
|       | 400  | 3.5                 | 0.06   | 63                           | 10    | 40    |
|       | 500  | 2.8                 | 0.06   | 57                           | 20    | 25    |
|       | 600  | 2.3                 | 0.06   | 72                           | 20    | 30    |
|       | 750  | 1.9                 | 0.06   | 72                           | 25    | 30    |
|       | 900  | 1.6                 | 0.06   | 73                           | 30    | 30    |
|       | 1200 | 1.2                 | 0.06   | 65                           | 30    | 40    |
|       | 1500 | 0.9                 | 0.06   | 73                           | 50    | 30    |
|       | 1800 | 0.8                 | 0.06   | 73                           | 60    | 30    |
|       | 2400 | 0.6                 | 0.06   | 65                           | 60    | 40    |
|       | 3200 | 0.4                 | 0.06   | 65                           | 80    | 40    |
| 30/50 | 300  | 4.7                 | 0.18   | 142                          | 10    | 30    |
|       | 400  | 3.5                 | 0.12   | 127                          | 10    | 40    |
|       | 500  | 2.8                 | 0.09   | 123                          | 10    | 50    |
|       | 600  | 2.3                 | 0.09   | 143                          | 20    | 30    |
|       | 750  | 1.9                 | 0.09   | 148                          | 25    | 30    |
|       | 900  | 1.6                 | 0.06   | 141                          | 30    | 30    |
|       | 1200 | 1.2                 | 0.06   | 118                          | 30    | 40    |
|       | 1500 | 0.9                 | 0.06   | 139                          | 50    | 30    |
|       | 1800 | 0.8                 | 0.06   | 155                          | 60    | 30    |
|       | 2400 | 0.6                 | 0.06   | 124                          | 60    | 40    |
|       | 3000 | 0.5                 | 0.06   | 120                          | 60    | 50    |
| 30/63 | 300  | 4.7                 | 0.22   | 210                          | 7.5   | 40    |
|       | 400  | 3.5                 | 0.18   | 222                          | 10    | 40    |
|       | 500  | 2.8                 | 0.18   | 205                          | 10    | 50    |
|       | 600  | 2.3                 | 0.12   | 208                          | 15    | 40    |
|       | 750  | 1.9                 | 0.12   | 216                          | 15    | 50    |
|       | 900  | 1.6                 | 0.09   | 200                          | 15    | 60    |
|       | 1200 | 1.2                 | 0.09   | 236                          | 30    | 40    |
|       | 1500 | 0.9                 | 0.06   | 204                          | 30    | 50    |
|       | 1800 | 0.8                 | 0.06   | 202                          | 30    | 60    |
|       | 2400 | 0.6                 | 0.06   | 220                          | 60    | 40    |
|       | 3000 | 0.5                 | 0.06   | 223                          | 60    | 50    |

| Size   | i    | $n_2(\text{r/min})$ | $kW_1$ | $M_2(\text{N}\cdot\text{m})$ | $i_1$ | $i_2$ |
|--------|------|---------------------|--------|------------------------------|-------|-------|
| 40/75  | 300  | 4.7                 | 0.37   | 405                          | 10    | 30    |
|        | 400  | 3.5                 | 0.25   | 336                          | 10    | 40    |
|        | 500  | 2.8                 | 0.25   | 307                          | 10    | 50    |
|        | 600  | 2.3                 | 0.18   | 362                          | 20    | 30    |
|        | 750  | 1.9                 | 0.18   | 391                          | 25    | 30    |
|        | 900  | 1.6                 | 0.12   | 325                          | 30    | 30    |
|        | 1200 | 1.2                 | 0.12   | 359                          | 30    | 40    |
|        | 1500 | 0.9                 | 0.09   | 360                          | 50    | 30    |
|        | 1800 | 0.8                 | 0.09   | 404                          | 60    | 30    |
|        | 2400 | 0.6                 | 0.06   | 330                          | 60    | 40    |
|        | 3000 | 0.5                 | 0.06   | 301                          | 60    | 50    |
| 40/90  | 300  | 4.7                 | 0.37   | 402                          | 7.5   | 40    |
|        | 400  | 3.5                 | 0.37   | 523                          | 10    | 40    |
|        | 500  | 2.8                 | 0.37   | 550                          | 10    | 50    |
|        | 600  | 2.3                 | 0.37   | 605                          | 15    | 40    |
|        | 750  | 1.9                 | 0.25   | 538                          | 15    | 50    |
|        | 900  | 1.6                 | 0.25   | 533                          | 15    | 60    |
|        | 1200 | 1.2                 | 0.18   | 629                          | 30    | 40    |
|        | 1500 | 0.9                 | 0.18   | 588                          | 30    | 50    |
|        | 1800 | 0.8                 | 0.12   | 492                          | 30    | 60    |
|        | 2400 | 0.6                 | 0.12   | 625                          | 60    | 40    |
|        | 3000 | 0.5                 | 0.09   | 548                          | 60    | 50    |
| 50/110 | 300  | 4.7                 | 0.75   | 879                          | 10    | 30    |
|        | 400  | 3.5                 | 0.75   | 1013                         | 10    | 40    |
|        | 500  | 2.8                 | 0.55   | 984                          | 10    | 50    |
|        | 600  | 2.3                 | 0.55   | 1062                         | 15    | 40    |
|        | 750  | 1.9                 | 0.55   | 1128                         | 25    | 30    |
|        | 900  | 1.6                 | 0.37   | 1079                         | 30    | 30    |
|        | 1200 | 1.2                 | 0.25   | 943                          | 30    | 40    |
|        | 1500 | 0.9                 | 0.25   | 1064                         | 50    | 30    |
|        | 1800 | 0.8                 | 0.25   | 1075                         | 60    | 30    |
|        | 2400 | 0.6                 | 0.18   | 1001                         | 60    | 40    |
|        | 3000 | 0.5                 | 0.12   | 884                          | 60    | 50    |
| 63/130 | 300  | 4.7                 | 1.50   | 1789                         | 10    | 30    |
|        | 400  | 3.5                 | 1.00   | 1519                         | 10    | 40    |
|        | 500  | 2.8                 | 1.00   | 1629                         | 10    | 50    |
|        | 600  | 2.3                 | 0.75   | 1631                         | 15    | 40    |
|        | 750  | 1.9                 | 0.75   | 1804                         | 25    | 30    |
|        | 900  | 1.6                 | 0.75   | 1826                         | 30    | 30    |
|        | 1200 | 1.2                 | 0.55   | 1705                         | 30    | 40    |
|        | 1500 | 0.9                 | 0.37   | 1674                         | 50    | 30    |
|        | 1800 | 0.8                 | 0.37   | 1698                         | 60    | 30    |
|        | 2400 | 0.6                 | 0.25   | 1624                         | 60    | 40    |
|        | 3000 | 0.5                 | 0.25   | 1548                         | 60    | 50    |
| 63/150 | 200  | 7                   | 1.5    | 1317                         | 10    | 20    |
|        | 250  | 5.6                 | 1.5    | 1602                         | 10    | 25    |
|        | 300  | 4.7                 | 1.5    | 1860                         | 10    | 30    |
|        | 400  | 3.5                 | 1.5    | 2208                         | 10    | 40    |
|        | 500  | 2.8                 | 1.1    | 1893                         | 20    | 25    |
|        | 600  | 2.3                 | 1.1    | 2242                         | 20    | 30    |
|        | 750  | 1.9                 | 0.75   | 1783                         | 25    | 30    |
|        | 900  | 1.6                 | 0.75   | 1994                         | 30    | 30    |
|        | 1200 | 1.2                 | 0.75   | 2680                         | 30    | 40    |
|        | 1500 | 0.9                 | 0.75   | 2700                         | 50    | 30    |
|        | 1800 | 0.8                 | 0.37   | 1775                         | 60    | 30    |
|        | 2400 | 0.6                 | 0.37   | 2141                         | 60    | 40    |
|        | 3000 | 0.5                 | 0.25   | 1713                         | 60    | 50    |



## MCFN ( $n_1=1400\text{r/m}$ )

### Performance parameter of Double MCFN ( worm-gear ) Series

| Size  |            | Ratio          | Input Power | Output              |                              |
|-------|------------|----------------|-------------|---------------------|------------------------------|
|       |            |                |             | $n_2(\text{r/min})$ | $M_2(\text{N}\cdot\text{m})$ |
| 40/63 | 75(3x25)   | 0.12           | 0.18        | 18.7                | 42                           |
|       |            |                |             |                     | 49                           |
|       | 90(3x30)   | 0.12           | 0.18        | 15.6                | 45                           |
|       |            |                |             |                     | 61                           |
|       | 120(3x40)  | 0.12           | 0.18        | 11.7                | 50                           |
|       |            |                |             |                     | 52                           |
|       | 150(3x50)  | 0.12           | 0.12        | 9.3                 | 46                           |
|       | 180(3x60)  |                |             | 7.8                 | 40                           |
| 50/   | 240(3x80)  |                |             | 5.8                 | 36                           |
|       | 300(3x100) |                |             | 4.7                 | 36                           |
|       | 63         | 75(3x25)       | 0.18        | 18.7                | 62                           |
|       | 71         | 79.3(3.17x25)  | 0.25        | 17.7                | 91                           |
|       | 63         | 90(3x30)       | 0.18        | 15.6                | 69                           |
|       | 71         | 95.1(3.17x30)  | 0.25        | 14.7                | 102                          |
|       | 63         | 120(3x40)      | 0.18        | 11.7                | 85                           |
|       | 71         | 126.8(3.17x40) | 0.25        | 11                  | 100                          |
|       | 63         | 150(3x50)      | 0.12        | 9.3                 | 66                           |
|       |            |                | 0.18        |                     | 89                           |
|       |            | 180(3x60)      | 0.12        | 7.8                 | 74                           |
|       |            |                | 0.18        |                     | 88                           |
|       |            | 240(3x80)      | 0.12        | 5.8                 | 78                           |
|       |            |                | 0.18        |                     | 76                           |
|       |            | 300(3x100)     | 0.12        | 4.7                 | 65                           |
|       |            |                | 0.18        |                     | 65                           |
| 63/   | 71         | 79.3(3.17x25)  | 0.25        | 17.7                | 94                           |
|       |            |                | 0.37        |                     | 139                          |
|       |            | 95.1(3.17x30)  | 0.25        | 14.7                | 103                          |
|       |            |                | 0.37        |                     | 153                          |
|       |            | 126.8(3.17x40) | 0.25        | 11                  | 129                          |
|       | 63         | 150(3x50)      | 0.18        | 9.3                 | 101                          |
|       |            |                | 0.25        |                     | 148                          |
|       |            | 158.5(3.17x50) | 0.37        | 8.8                 | 176                          |
|       |            |                | 0.18        |                     | 115                          |
|       |            | 190.2(3.17x60) | 0.25        | 7.4                 | 151                          |
|       | 63         | 240(3x80)      | 0.12        | 5.8                 | 90                           |
|       |            |                | 0.18        |                     | 136                          |
|       |            | 253.6(3.17x80) | 0.25        | 5.5                 | 139                          |
|       |            |                | 0.12        |                     | 101                          |
|       |            | 300(3x100)     | 0.18        | 4.7                 | 121                          |
| 75/   | 80         | 75(3x25)       | 0.55        | 18.7                | 201                          |
|       |            |                | 0.75        |                     | 247                          |
|       |            |                | 0.92        |                     | 269                          |
|       | 71         | 79.3(3.17x25)  | 0.37        | 17.7                | 143                          |
|       |            |                | 0.55        |                     | 225                          |
|       |            |                | 0.75        |                     | 307                          |
|       | 80         | 90(3x30)       | 0.92        | 15.6                | 300                          |
|       |            |                | 0.37        |                     | 160                          |
|       |            |                | 0.55        |                     | 278                          |
|       | 71         | 126.8(3.17x40) | 0.37        | 11                  | 198                          |
|       |            |                | 0.55        |                     | 260                          |
|       |            |                | 0.25        |                     | 156                          |
|       | 71         | 158.5(3.17x50) | 0.37        | 8.8                 | 231                          |
|       |            |                | 0.37        |                     | 231                          |

| Size  |    | Ratio          | Input Power | Output              |                              |
|-------|----|----------------|-------------|---------------------|------------------------------|
|       |    |                |             | $n_2(\text{r/min})$ | $M_2(\text{N}\cdot\text{m})$ |
| 75/71 |    | 190.2(3.17x60) | 0.25        | 7.4                 | 178                          |
|       |    |                | 0.37        |                     | 236                          |
|       |    | 253.6(3.17x80) | 0.25        | 5.5                 | 208                          |
|       |    |                |             | 4.4                 | 214                          |
| 90/   | 80 | 90(3x30)       | 0.55        | 15.6                | 235                          |
|       |    |                | 0.75        |                     | 320                          |
|       |    | 120(3x40)      | 0.55        | 11.7                | 291                          |
|       |    |                | 0.75        |                     | 397                          |
|       |    | 150(3x50)      | 0.55        | 9.3                 | 347                          |
|       |    |                | 0.75        |                     | 426                          |
|       |    | 180(3x60)      | 0.55        | 7.8                 | 390                          |
|       |    |                | 0.75        |                     | 425                          |
|       | 71 | 190.2(3.17x60) | 0.37        | 7.4                 | 278                          |
|       | 80 | 240(3x80)      | 0.55        | 5.8                 | 374                          |
|       |    |                |             |                     | 332                          |
|       | 71 | 253.6(3.17x80) | 0.37        | 4.4                 | 345                          |
|       |    |                |             |                     | 384                          |
| 110/  | 90 | 72.6(2.42x30)  | 1.1         | 19.3                | 524                          |
|       |    |                | 1.5         |                     | 629                          |
|       |    |                | 1.8         |                     | 293                          |
|       | 80 | 75(3x25)       | 0.75        | 18.7                | 498                          |
|       |    |                | 1.1         |                     | 679                          |
|       |    |                | 1.5         |                     | 815                          |
|       | 90 | 96.8(2.42x40)  | 1.1         | 14.5                | 587                          |
|       |    |                | 1.5         |                     | 801                          |
|       |    |                | 1.8         |                     | 768                          |
|       | 80 | 121(2.42x50)   | 1.1         | 11.6                | 673                          |
|       |    |                | 1.5         |                     | 733                          |
|       |    |                | 1.8         |                     | 496                          |
|       | 90 | 145.2(2.42x60) | 0.75        | 9.3                 | 417                          |
|       |    |                | 0.55        |                     | 569                          |
|       |    |                | 0.75        |                     | 648                          |
| 130/  | 80 | 193.6(2.42x80) | 1.1         | 7.2                 | 503                          |
|       |    |                | 0.55        |                     | 617                          |
|       |    |                | 0.55        |                     | 585                          |
|       | 90 | 240(3x80)      | 0.55        | 5.8                 | 390                          |
|       |    |                | 0.75        |                     | 531                          |
|       |    | 300(3x100)     | 0.55        | 4.7                 | 638                          |
|       |    |                | 0.75        |                     | 815                          |
|       |    | 72.6(2.42x30)  | 1.1         | 19.3                | 596                          |
|       |    |                | 1.5         |                     | 813                          |
|       |    | 96.8(2.42x40)  | 1.1         | 14.5                | 976                          |
|       |    |                | 1.5         |                     | 673                          |
|       | 80 | 121(2.42x50)   | 1.1         | 11.6                | 917                          |
|       |    |                | 1.5         |                     | 1101                         |
|       |    | 145.2(2.42x60) | 1.1         | 9.6                 | 826                          |
|       |    |                | 1.5         |                     | 1013                         |
|       |    | 193.6(2.42x80) | 1.1         | 7.2                 | 698                          |
|       |    |                | 1.5         |                     | 848                          |
| 80    |    | 242(2.42x100)  | 0.75        | 5.8                 | 585                          |
|       |    |                | 0.75        |                     | 797                          |



## MCNUD

### Performance parameter of MCNUD ( worm-variator ) Series

| Input                                     | Size     | Ratio | Output       |            |
|-------------------------------------------|----------|-------|--------------|------------|
|                                           |          |       | $n_2(r/min)$ | $M_2(N.m)$ |
| $P_1 = 0.18kw$<br>4p<br>$n_1 = 1400r/min$ | 40/0.18  | 7.5   | 117~22.7     | 9~18       |
|                                           | 40/0.18  | 10    | 88~17        | 12~23      |
|                                           | 40/0.18  | 15    | 58.7~11.3    | 17~32      |
|                                           | 40/0.18  | 20    | 44~8.5       | 22~40      |
|                                           | 40/0.18  | 25    | 35.2~6.8     | 27~47      |
|                                           | 40/0.18  | 30    | 29.3~5.7     | 30~51      |
|                                           | 40/0.18  | 40    | 22~4.3       | 37~62      |
|                                           | 50/0.18  | 40    | 22~4.3       | 38~63      |
|                                           | 50/0.18  | 50    | 17.6~3.4     | 43~60      |
|                                           | 50/0.18  | 50    | 17.6~3.4     | 44~73      |
|                                           | 50/0.18  | 60    | 14.7~2.8     | 50~80      |
|                                           | 50/0.18  | 80    | 11~2.1       | 59~82      |
| $P_1 = 0.37kw$<br>4p<br>$n_1 = 1400r/min$ | 50/0.18  | 100   | 8.8~1.7      | 66~79      |
|                                           | 50/0.37  | 7.5   | 133~26.7     | 19~36      |
|                                           | 50/0.37  | 10    | 100~20       | 25~47      |
|                                           | 50/0.37  | 15    | 66.7~13.3    | 36~65      |
|                                           | 50/0.37  | 20    | 50~10        | 46~82      |
|                                           | 50/0.37  | 25    | 40~8         | 55~97      |
|                                           | 50/0.37  | 30    | 33.3~6.7     | 61~107     |
|                                           | 50/0.37  | 40    | 25~5         | 76~124     |
|                                           | 63/0.37  | 40    | 25~5         | 79~134     |
|                                           | 50/0.37  | 50    | 20~4         | 89~120     |
|                                           | 63/0.37  | 50    | 20~4         | 92~155     |
|                                           | 63/0.37  | 60    | 16.7~3.3     | 104~173    |
| $P_1 = 0.55kw$<br>4p<br>$n_1 = 1400r/min$ | 63/0.37  | 80    | 12.5~2.5     | 125~173    |
|                                           | 63/0.37  | 100   | 10~2         | 139~150    |
|                                           | 63/0.55  | 7.5   | 133~26.7     | 26~49      |
|                                           | 63/0.55  | 10    | 100~20       | 34~63      |
|                                           | 63/0.55  | 15    | 66.7~13.3    | 48~88      |
|                                           | 63/0.55  | 20    | 50~10        | 62~112     |
|                                           | 63/0.55  | 25    | 40~8         | 75~133     |
|                                           | 63/0.55  | 30    | 33.3~6.7     | 81~146     |
|                                           | 63/0.55  | 40    | 25~5         | 105~179    |
|                                           | 63/0.55  | 50    | 20~4         | 123~207    |
|                                           | 75/0.55  | 50    | 20~4         | 129~216    |
|                                           | 75/0.55  | 60    | 16.7~3.3     | 146~242    |
| $P_1 = 0.75kw$<br>4p<br>$n_1 = 1400r/min$ | 75/0.55  | 80    | 12.5~2.5     | 176~250    |
|                                           | 90/0.55  | 80    | 12.5~2.5     | 189~309    |
|                                           | 90/0.55  | 100   | 10~2         | 218~350    |
|                                           | 63/0.75  | 7.5   | 133~26.7     | 39~73      |
|                                           | 63/0.75  | 10    | 100~20       | 51~94      |
|                                           | 63/0.75  | 15    | 66.7~13.3    | 72~132     |
|                                           | 63/0.75  | 20    | 50~10        | 92~168     |
|                                           | 63/0.75  | 25    | 40~8         | 112~199    |
|                                           | 63/0.75  | 30    | 33.3~6.7     | 126~219    |
|                                           | 63/0.75  | 40    | 25~5         | 156~232    |
|                                           | 63/0.75  | 50    | 20~4         | 185~310    |
|                                           | 75/0.75  | 50    | 20~4         | 192~320    |
| $P_1 = 1.1kw$<br>4p<br>$n_1 = 1400r/min$  | 75/0.75  | 60    | 16.7~3.3     | 219~300    |
|                                           | 90/0.75  | 60    | 16.7~3.3     | 230~389    |
|                                           | 90/0.75  | 80    | 12.5~2.5     | 265~428    |
|                                           | 110/0.75 | 80    | 12.5~2.5     | 302~503    |
|                                           | 90/0.75  | 100   | 10~2         | 303~410    |
|                                           | 110/0.75 | 100   | 10~2         | 348~575    |
|                                           | 75/1.1   | 7.5   | 133~26.7     | 59~111     |
|                                           | 75/1.1   | 10    | 100~20       | 77~144     |
|                                           | 90/1.1   | 10    | 100~20       | 78~146     |
|                                           | 75/1.1   | 15    | 66.7~13.3    | 110~203    |
|                                           | 90/1.1   | 15    | 66.7~13.3    | 113~208    |
|                                           | 75/1.1   | 20    | 50~10        | 142~258    |

| Input                                    | Size    | Ratio | Output       |            |
|------------------------------------------|---------|-------|--------------|------------|
|                                          |         |       | $n_2(r/min)$ | $M_2(N.m)$ |
| $P_1 = 1.1kw$<br>4p<br>$n_1 = 1400r/min$ | 90/1.1  | 50    | 20~4         | 304~517    |
|                                          | 110/1.1 | 50    | 20~4         | 320~550    |
|                                          | 110/1.1 | 60    | 16.7~3.3     | 368~625    |
|                                          | 130/1.1 | 60    | 16.7~3.3     | 373~623    |
|                                          | 110/1.1 | 80    | 12.5~2.5     | 455~754    |
|                                          | 130/1.1 | 80    | 12.5~2.5     | 460~749    |
|                                          | 110/1.1 | 100   | 10~2         | 522~170    |
|                                          | 130/1.1 | 100   | 10~2         | 531~868    |
|                                          | 75/1.5  | 7.5   | 133~26.7     | 78~148     |
|                                          | 90/1.5  | 7.5   | 133~26.7     | 77~150     |
|                                          | 75/1.5  | 10    | 100~20       | 102~192    |
|                                          | 90/1.5  | 10    | 100~20       | 104~195    |
| $P_1 = 1.5kw$<br>4p<br>$n_1 = 1400r/min$ | 75/1.5  | 15    | 66.7~13.3    | 147~270    |
|                                          | 90/1.5  | 15    | 66.7~13.3    | 150~277    |
|                                          | 75/1.5  | 20    | 50~10        | 190~344    |
|                                          | 90/1.5  | 20    | 50~10        | 194~355    |
|                                          | 75/1.5  | 25    | 40~8         | 229~330    |
|                                          | 90/1.5  | 25    | 40~8         | 236~427    |
|                                          | 75/1.5  | 30    | 33.3~6.7     | 260~390    |
|                                          | 90/1.5  | 30    | 33.3~6.7     | 270~474    |
|                                          | 75/1.5  | 40    | 25~5         | 327~360    |
|                                          | 90/1.5  | 40    | 25~5         | 341~589    |
|                                          | 90/1.5  | 50    | 20~4         | 406~560    |
|                                          | 110/1.5 | 50    | 20~4         | 426~733    |
| $P_1 = 2.2kw$<br>4p<br>$n_1 = 1400r/min$ | 110/1.5 | 60    | 16.7~3.3     | 490~833    |
|                                          | 130/1.5 | 60    | 16.7~3.3     | 498~831    |
|                                          | 130/1.5 | 80    | 12.5~2.5     | 614~999    |
|                                          | 130/1.5 | 100   | 10~2         | 696~1100   |
|                                          | 110/2.2 | 7.5   | 133~26.7     | 120~226    |
|                                          | 110/2.2 | 10    | 100~20       | 157~294    |
|                                          | 110/2.2 | 15    | 66.7~13.3    | 228~418    |
|                                          | 110/2.2 | 20    | 50~10        | 298~549    |
|                                          | 110/2.2 | 25    | 40~8         | 364~664    |
|                                          | 110/2.2 | 30    | 33.3~6.7     | 413~717    |
|                                          | 110/2.2 | 40    | 25~5         | 533~931    |
|                                          | 130/2.2 | 40    | 25~5         | 542~932    |
| $P_1 = 3.0kw$<br>4p<br>$n_1 = 1400r/min$ | 130/2.2 | 50    | 20~4         | 648~1097   |
|                                          | 130/2.2 | 60    | 16.7~3.3     | 746~1246   |
|                                          | 130/2.2 | 80    | 12.5~2.5     | 921~1499   |
|                                          | 130/2.2 | 100   | 10~2         | 1040~1100  |
|                                          | 110/3.0 | 7.5   | 133~26.7     | 160~302    |
|                                          | 130/3.0 | 7.5   | 133~26.7     | 160~301    |
|                                          | 110/3.0 | 10    | 100~20       | 210~392    |
|                                          | 130/3.0 | 10    | 100~20       | 211~395    |
|                                          | 110/3.0 | 15    | 66.7~13.3    | 304~558    |
|                                          | 130/3.0 | 15    | 66.7~13.3    | 307~563    |
|                                          | 110/3.0 | 20    | 50~10        | 398~732    |
|                                          | 130/3.0 | 20    | 50~10        | 402~733    |
| $P_1 = 4.0kw$<br>4p<br>$n_1 = 1400r/min$ | 110/3.0 | 25    | 40~8         | 485~885    |
|                                          | 130/3.0 | 25    | 40~8         | 490~885    |
|                                          | 110/3.0 | 30    | 33.3~6.7     | 547~956    |
|                                          | 130/3.0 | 30    | 33.3~6.7     | 562~973    |
|                                          | 110/3.0 | 40    | 25~5         | 711~1030   |
|                                          | 130/3.0 | 40    | 25~5         | 720~1242   |
|                                          | 130/3.0 | 50    | 20~4         | 864~1463   |
|                                          | 110/4.0 | 7.5   | 133~26.7     | 213~402    |
|                                          | 130/4.0 | 7.5   | 133~26.7     | 214~401    |
|                                          | 110/4.0 | 10    | 100~20       | 279~523    |
|                                          | 130/4.0 | 10    | 100~20       | 281~527    |
|                                          | 110/4.0 | 15    | 66.7~13.3    | 405~744    |



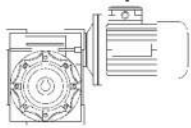
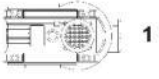
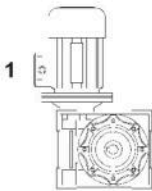
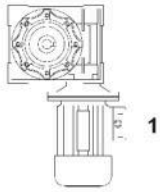
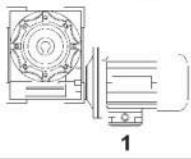
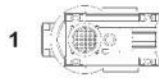
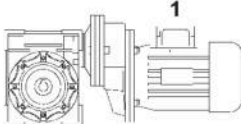
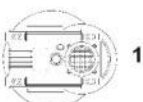
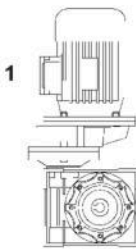
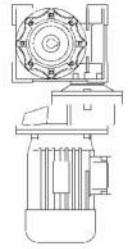
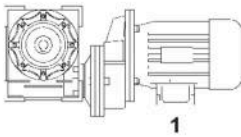
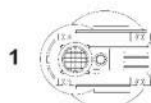
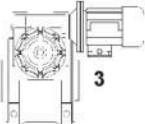
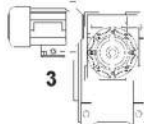
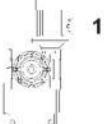
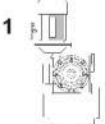
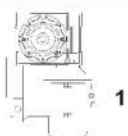
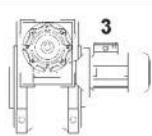
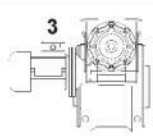
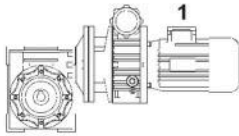
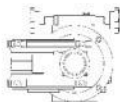
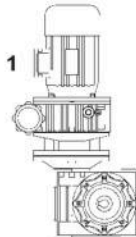
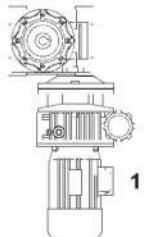
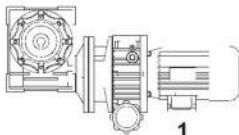
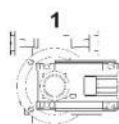
## Specifications of lubricants

|                            |  |  |  |  |  |  |  |  |  |               |
|----------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|---------------|
|                            | °C -50 0 +50 +100                                                                 | ISO                                                                               | SHELL                                                                             | AGIP                                                                              | ESSO                                                                              | MOBIL                                                                              | CASTROL                                                                             | BP                                                                                  | GEMER                                                                               |               |
| MCNDK 025~090<br>PC063~090 | -25 +50                                                                           | VG320                                                                             | Tivela<br>OIL S320                                                                | Telium<br>VSF320                                                                  | S220                                                                              | Glygoyle<br>30                                                                     | Alphasyn<br>PG320                                                                   | Energol<br>SG-XP320                                                                 | WA460                                                                               | Synthetic oil |
| MCNDK 110~130              | -5 +40                                                                            | VG460                                                                             | Omala<br>OIL460                                                                   | Blasia<br>460                                                                     | Spartan<br>EP460                                                                  | Mobilgear<br>634                                                                   | Alpha<br>MAX<br>460                                                                 | Energol<br>SG-XP460                                                                 | WA460                                                                               | Mineral oil   |
|                            | -15 +25                                                                           | VG220                                                                             | Omala<br>OIL220                                                                   | Blasia<br>220                                                                     | Spartan<br>EP220                                                                  | Mobilgear<br>630                                                                   | Alpha<br>MAX<br>220                                                                 | Energol<br>SG-XP220                                                                 | WA460                                                                               |               |
| PC                         | -15 +50                                                                           | VG320                                                                             | Tivela<br>OIL S320                                                                | Telium<br>VSF320                                                                  | S220                                                                              | Glygoyle<br>30                                                                     | Alphasyn<br>PG320                                                                   | Snergol<br>SG-XP320                                                                 | CKC150                                                                              | Synthetic oil |
| VSA                        | -25 +40                                                                           | VG32                                                                              | A.T.F.DXRON                                                                       | A.T.F.DXRON                                                                       | A.T.F.DXRON                                                                       | A.T.F. 220                                                                         | TQ.DXRON II                                                                         | Autran DX                                                                           | Ub3                                                                                 | Mineral oil   |

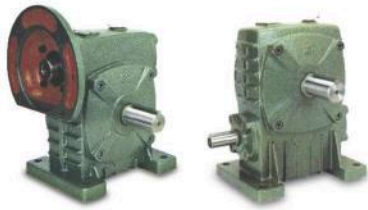
## Quantity of oil in litres

| MCNDK | 25      | 30      | 40      | 50      | 63    | 75    | 90    | 110  | 130  | PC | 63   | 71   | 80   | 90   |
|-------|---------|---------|---------|---------|-------|-------|-------|------|------|----|------|------|------|------|
| B3    | 0.023   | 0.05    | 0.1     | 0.15    | 0.3   | 0.5   | 1     | 3    | 4.5  |    | 0.05 | 0.07 | 0.15 | 0.16 |
| B8    |         |         |         |         |       |       |       | 2.2  | 3.3  |    |      |      |      |      |
| B6-B7 |         |         |         |         |       |       |       | 2.5  | 3.5  |    |      |      |      |      |
| V5    |         |         |         |         |       |       |       | 3    | 4.5  |    |      |      |      |      |
| V6    |         |         |         |         |       |       |       | 2.2  | 3.3  |    |      |      |      |      |
| VSA   | VSA0.18 | VSA0.37 | VSA0.55 | VSA0.75 | VS1.1 | VS1.5 | VS2.2 | VS3  | VS4  |    |      |      |      |      |
| B3    | 0.13    | 0.15    | 0.33    | 0.33    | 0.8   | 0.8   | 1.2   | 1.2  | 1.2  |    |      |      |      |      |
| B8    |         |         |         |         |       |       |       |      |      |    |      |      |      |      |
| B6-B7 |         |         |         |         |       |       |       |      |      |    |      |      |      |      |
| V5    | 0.2     | 0.25    | 0.45    | 0.45    | 1     | 1     | 1.2   | 1.2  | 1.2  |    |      |      |      |      |
| V6    | 0.3     | 0.4     | 0.85    | 0.85    | 1.4   | 1.4   | 2.15  | 2.15 | 2.15 |    |      |      |      |      |

## Mounting positions

| MCNDK                                                                               |                                                                                     |                                                                                      |                                                                                       |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| MCNDK - B3                                                                          | B6                                                                                  | V5                                                                                   | V6                                                                                    |
|    |    |    |    |
| B8                                                                                  | B7                                                                                  |                                                                                      |                                                                                       |
|    |    |                                                                                      |                                                                                       |
| MCFNDK                                                                              |                                                                                     |                                                                                      |                                                                                       |
| MCFNDK - B3                                                                         | B6                                                                                  | V5                                                                                   | V6                                                                                    |
|    |    |   |   |
| B8                                                                                  | B7                                                                                  |                                                                                      |                                                                                       |
|  |  |                                                                                      |                                                                                       |
| MCENDK / WORM - WORM                                                                |                                                                                     |                                                                                      |                                                                                       |
| AS1                                                                                 | AS2                                                                                 | VS1                                                                                  | VS2                                                                                   |
|  |  |  |  |
| PS1                                                                                 | PS2                                                                                 | BS1                                                                                  | BS2                                                                                   |
|  |  |  |  |
| MBL - WORM                                                                          |                                                                                     |                                                                                      |                                                                                       |
| MCNUD - B3                                                                          | B6                                                                                  | V5                                                                                   | V6                                                                                    |
|  |  |  |  |
| B8                                                                                  | B7                                                                                  |                                                                                      |                                                                                       |
|  |  |                                                                                      |                                                                                       |





**WORM GEAR REDUCER**

Ratio 1:10 - 1:60



**HELICAL GEAR MOTOR**

Ratio 1:5 - 1:200



**VARIABLE SPEED GEAR MOTOR**

Speed 200 - 1000 RPM



**HELICAL GEAR**

Ratio 1:5 - 1:264



**HYPOID GEAR MOTOR**

Ratio 1:5 - 1:240



**AC & DC RIGHT ANGLE GEAR MOTOR**

Ratio 1:7.5 - 1:180



**RV CYCLOIDAL PIN WHEEL REDUCER**

Ratio 1:43 - 1:192.4



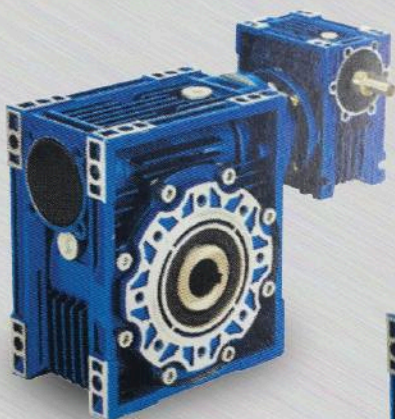
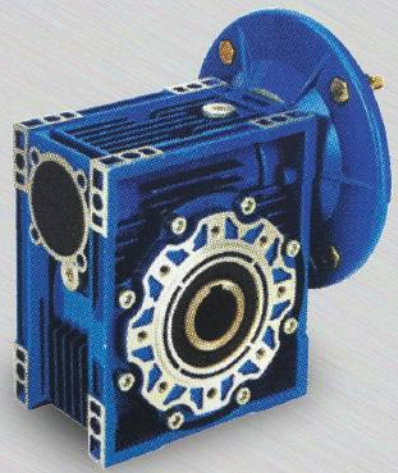
**PLANETARY GEAR BOX**

Single Stage Ratio 1:3 - 1:10

Two Stage Ratio 1:9 - 1:64

Three Stage Ratio 1:60 - 1:516





**MaxLoyd**  
moving power

